



US012408291B2

(12) **United States Patent**
Yeung et al.

(10) **Patent No.:** **US 12,408,291 B2**

(45) **Date of Patent:** **Sep. 2, 2025**

(54) **ENCLOSURE ASSEMBLY FOR ENHANCED COOLING OF DIRECT DRIVE UNIT AND RELATED METHODS**

(56) **References Cited**

U.S. PATENT DOCUMENTS

(71) Applicant: **BJ Energy Solutions, LLC**, Houston, TX (US)

1,716,049 A 6/1929 Greve
1,726,633 A 9/1929 Smith
(Continued)

(72) Inventors: **Tony Yeung**, Houston, TX (US);
Ricardo Rodriguez-Ramon, Houston, TX (US); **Guillermo Rodriguez**, Houston, TX (US)

FOREIGN PATENT DOCUMENTS

AU 9609498 7/1999
AU 737970 9/2001

(Continued)

(73) Assignee: **BJ Energy Solutions, LLC**, The Woodlands, TX (US)

OTHER PUBLICATIONS

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 329 days.

Rigmaster Machinery Ltd., Model: 2000 RMP-6-PLEX, brochure, downloaded at https://www.rigmastermachinery.com/_files/ugd/431e62_eaec77c9fe54af8b13d08396072da67.pdf.

(Continued)

(21) Appl. No.: **18/327,599**

(22) Filed: **Jun. 1, 2023**

Primary Examiner — Mandeep S Buttar

(74) *Attorney, Agent, or Firm* — Norton Rose Fulbright US LLP

(65) **Prior Publication Data**

US 2023/0320025 A1 Oct. 5, 2023

Related U.S. Application Data

(60) Continuation of application No. 18/096,927, filed on Jan. 13, 2023, now Pat. No. 11,723,171, which is a (Continued)

(51) **Int. Cl.**
H05K 7/20 (2006.01)
E21B 43/26 (2006.01)

(52) **U.S. Cl.**
CPC **H05K 7/20154** (2013.01); **E21B 43/26** (2013.01); **H05K 7/20145** (2013.01); **H05K 7/20209** (2013.01)

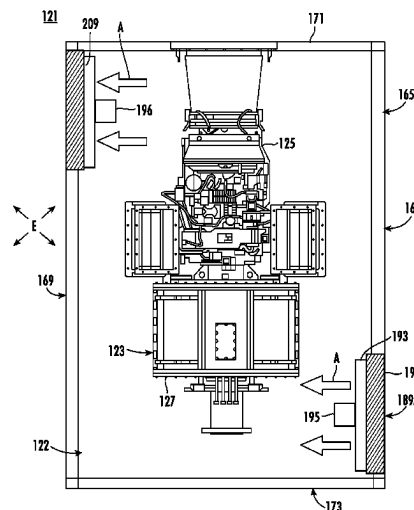
(58) **Field of Classification Search**
CPC .. H05K 7/20145; H05K 7/20154; F16N 7/00; H02K 7/1823; H02K 9/06; F04B 17/05;

(Continued)

(57) **ABSTRACT**

Embodiments of an enclosure assembly to enhance cooling of a hydraulic fracturing direct drive unit (DDU) during operation are included. The enclosure assembly may include an enclosure body extending at least partially around an enclosure space to house the DDU for driving a fluid pump. The enclosure assembly may include one or more heat exchanger assemblies connected to the enclosure body for cooling a process fluid associated with one or more of the DDU and the fluid pump, and which may be configured to draw air into the enclosure space from and external environment, toward one or more radiator assemblies to cool the process fluid, and along an airflow path through the enclosure space. One or more outlet fan assemblies may be operative to discharge air from the enclosure space to the external environment to maintain a desired temperature of the enclosure space.

16 Claims, 12 Drawing Sheets



Related U.S. Application Data

continuation of application No. 17/444,485, filed on Aug. 5, 2021, now Pat. No. 11,627,683, which is a continuation of application No. 17/356,063, filed on Jun. 23, 2021, now Pat. No. 11,129,295, which is a division of application No. 17/302,039, filed on Apr. 22, 2021, now Pat. No. 11,109,508.

- (60) Provisional application No. 62/705,042, filed on Jun. 9, 2020, provisional application No. 62/704,981, filed on Jun. 5, 2020.

(58) **Field of Classification Search**

CPC F04B 2203/11; F04B 51/00; F04B 15/02;
E21B 43/2607

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2,178,662	A	11/1939	Lars	4,269,569	A	5/1981	Hoover
2,427,638	A	9/1947	Vilter	4,311,395	A	1/1982	Douthitt et al.
2,498,229	A	2/1950	Adler	4,330,237	A	5/1982	Battah
2,535,703	A	12/1950	Smith et al.	4,341,508	A	7/1982	Rambin, Jr.
2,572,711	A	10/1951	Fischer	4,357,027	A	11/1982	Zeitlow
2,820,341	A	1/1958	Amann	4,383,478	A	5/1983	Jones
2,868,004	A	1/1959	Runde	4,402,504	A	9/1983	Christian
2,940,377	A	6/1960	Darnell et al.	4,430,047	A	2/1984	Ilg
2,947,141	A	8/1960	Russ	4,442,665	A	4/1984	Fick
2,956,738	A	10/1960	Rosenschold	4,457,325	A	7/1984	Green
3,068,796	A	12/1962	Pfluger et al.	4,470,771	A	9/1984	Hall et al.
3,191,517	A	6/1965	Solzman	4,483,684	A	11/1984	Black
3,257,031	A	6/1966	Dietz	4,505,650	A	3/1985	Hannett et al.
3,274,768	A	9/1966	Klein	4,574,880	A	3/1986	Handke
3,378,074	A	4/1968	Kiel	4,584,654	A	4/1986	Crane
3,382,671	A	5/1968	Ehni, III	4,620,330	A	11/1986	Izzi, Sr.
3,401,873	A	9/1968	Privon	4,672,813	A	6/1987	David
3,463,612	A	8/1969	Whitsel	4,754,607	A	7/1988	Mackay
3,496,880	A	2/1970	Wolff	4,782,244	A	11/1988	Wakimoto
3,550,696	A	12/1970	Kenneday	4,796,777	A	1/1989	Keller
3,560,053	A	2/1971	Orloff	4,869,209	A	9/1989	Young
3,586,459	A	6/1971	Zerlauth	4,913,625	A	4/1990	Gerlowski
3,632,222	A	1/1972	Cronstedt	4,983,259	A	1/1991	Duncan
3,656,582	A	4/1972	Alcock	4,990,058	A	2/1991	Eslinger
3,667,868	A	6/1972	Brunner	5,032,065	A	7/1991	Yamamuro
3,692,434	A	9/1972	Schnear	5,135,361	A	8/1992	Dion
3,739,872	A	6/1973	McNair	5,167,493	A	12/1992	Kobari
3,757,581	A	9/1973	Mankin	5,245,970	A	9/1993	Iwaszkiewicz et al.
3,759,063	A	9/1973	Bendall	5,291,842	A	3/1994	Sallstrom et al.
3,765,173	A	10/1973	Harris	5,326,231	A	7/1994	Pandeya
3,771,916	A	11/1973	Flanigan et al.	5,362,219	A	11/1994	Paul et al.
3,773,438	A	11/1973	Hall et al.	5,511,956	A	4/1996	Hasegawa
3,781,135	A	12/1973	Nickell	5,537,813	A	7/1996	Davis et al.
3,786,835	A	1/1974	Finger	5,553,514	A	9/1996	Walkowc
3,791,682	A	2/1974	Mitchell	5,560,195	A	10/1996	Anderson et al.
3,796,045	A	3/1974	Foster	5,586,444	A	12/1996	Fung
3,814,549	A	6/1974	Cronstedt	5,622,245	A	4/1997	Reik
3,820,922	A	6/1974	Buse et al.	5,626,103	A	5/1997	Haws et al.
3,847,511	A	11/1974	Cole	5,634,777	A	6/1997	Albertin
3,866,108	A	2/1975	Yannone	5,651,400	A	7/1997	Corts et al.
3,875,380	A	4/1975	Rankin	5,678,460	A	10/1997	Walkowc
3,963,372	A	6/1976	McLain et al.	5,717,172	A	2/1998	Griffin, Jr. et al.
4,010,613	A	3/1977	McInerney	5,720,598	A	2/1998	de Chizzelle
4,019,477	A	4/1977	Overton	5,761,084	A	6/1998	Edwards
4,031,407	A	6/1977	Reed	5,811,676	A	9/1998	Spalding et al.
4,050,862	A	9/1977	Buse	5,839,888	A	11/1998	Harrison
4,059,045	A	11/1977	McClain	5,846,062	A	12/1998	Yanagisawa et al.
4,086,976	A	5/1978	Holm et al.	5,875,744	A	3/1999	Vallejos
4,117,342	A	9/1978	Melley, Jr.	5,983,962	A	11/1999	Gerardot
4,173,121	A	11/1979	Yu	5,992,944	A	11/1999	Hara
4,204,808	A	5/1980	Reese et al.	6,041,856	A	3/2000	Thrasher et al.
4,209,079	A	6/1980	Marchal et al.	6,050,080	A	4/2000	Horner
4,209,979	A	7/1980	Woodhouse et al.	6,067,962	A	5/2000	Bartley et al.
4,222,229	A	9/1980	Uram	6,071,188	A	6/2000	O'Neill et al.
4,239,396	A	12/1980	Arribau et al.	6,074,170	A	6/2000	Bert et al.
				6,123,751	A	9/2000	Nelson et al.
				6,129,335	A	10/2000	Yokogi
				6,145,318	A	11/2000	Kaplan et al.
				6,230,481	B1	5/2001	Jahr
				6,279,309	B1	8/2001	Lawlor, II et al.
				6,321,860	B1	11/2001	Reddoch
				6,334,746	B1	1/2002	Nguyen et al.
				6,401,472	B2	6/2002	Pollrich
				6,530,224	B1	3/2003	Conchieri
				6,543,395	B2	4/2003	Green
				6,655,922	B1	12/2003	Flek
				6,669,453	B1	12/2003	Breeden
				6,765,304	B2	7/2004	Baten et al.
				6,786,051	B2	9/2004	Kristich et al.
				6,832,900	B2	12/2004	Leu
				6,851,514	B2	2/2005	Han et al.
				6,859,740	B2	2/2005	Stephenson et al.
				6,901,735	B2	6/2005	Lohn
				6,962,057	B2	11/2005	Kurokawa et al.
				7,007,966	B2	3/2006	Campion
				7,047,747	B2	5/2006	Tanaka
				7,065,953	B1	6/2006	Kopko
				7,143,016	B1	11/2006	Discenzo et al.
				7,222,015	B2	5/2007	Davis et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

7,281,519 B2	10/2007	Schroeder	9,206,667 B2	12/2015	Khvoshchev et al.
7,388,303 B2	6/2008	Seiver	9,212,643 B2	12/2015	Deliyski
7,404,294 B2	7/2008	Sundin	9,222,346 B1	12/2015	Walls
7,442,239 B2	10/2008	Armstrong et al.	9,324,049 B2	4/2016	Thomeer et al.
7,524,173 B2	4/2009	Cummins	9,341,055 B2	5/2016	Weightman et al.
7,545,130 B2	6/2009	Latham	9,346,662 B2	5/2016	Van Vliet et al.
7,552,903 B2	6/2009	Dunn et al.	9,366,114 B2	6/2016	Coli et al.
7,563,076 B2	7/2009	Brunet et al.	9,376,786 B2	6/2016	Numasawa
7,563,413 B2	7/2009	Naets et al.	9,394,829 B2	7/2016	Cabeen et al.
7,574,325 B2	8/2009	Dykstra	9,395,049 B2	7/2016	Vicknair et al.
7,581,379 B2	9/2009	Yoshida et al.	9,401,670 B2	7/2016	Minato et al.
7,594,424 B2	9/2009	Fazekas	9,410,410 B2	8/2016	Broussard et al.
7,614,239 B2	11/2009	Herzog et al.	9,410,546 B2	8/2016	Jaeger et al.
7,627,416 B2	12/2009	Batenburg et al.	9,429,078 B1	8/2016	Crowe et al.
7,677,316 B2	3/2010	Butler et al.	9,435,333 B2	9/2016	McCoy et al.
7,721,521 B2	5/2010	Kunkle et al.	9,488,169 B2	11/2016	Cochran et al.
7,730,711 B2	6/2010	Kunkle et al.	9,493,997 B2	11/2016	Liu et al.
7,779,961 B2	8/2010	Matte	9,512,783 B2	12/2016	Veilleux et al.
7,789,452 B2	9/2010	Dempsey et al.	9,534,473 B2	1/2017	Morris et al.
7,836,949 B2	11/2010	Dykstra	9,546,652 B2	1/2017	Yin
7,841,394 B2	11/2010	McNeel et al.	9,550,501 B2	1/2017	Ledbetter
7,845,413 B2	12/2010	Shampine et al.	9,556,721 B2	1/2017	Jang et al.
7,861,679 B2	1/2011	Lemke et al.	9,562,420 B2	2/2017	Morris et al.
7,886,702 B2	2/2011	Jerrell et al.	9,570,945 B2	2/2017	Fischer
7,900,724 B2	3/2011	Promersberger et al.	9,579,980 B2	2/2017	Cryer et al.
7,921,914 B2	4/2011	Bruins et al.	9,587,649 B2	3/2017	Oehring
7,938,151 B2	5/2011	Höckner	9,593,710 B2	3/2017	Laimboeck et al.
7,955,056 B2	6/2011	Pettersson	9,611,728 B2	4/2017	Oehring
7,980,357 B2	7/2011	Edwards	9,617,808 B2	4/2017	Liu et al.
8,056,635 B2	11/2011	Shampine et al.	9,638,101 B1	5/2017	Crowe et al.
8,083,504 B2	12/2011	Williams et al.	9,638,194 B2	5/2017	Wiegman et al.
8,099,942 B2	1/2012	Alexander	9,650,871 B2	5/2017	Oehring et al.
8,186,334 B2	5/2012	Ooyama	9,656,762 B2	5/2017	Kamath et al.
8,196,555 B2	6/2012	Ikeda et al.	9,689,316 B1	6/2017	Crom
8,202,354 B2	6/2012	Iijima	9,695,808 B2	7/2017	Giessbach et al.
8,316,936 B2	11/2012	Roddy et al.	9,739,130 B2	8/2017	Young
8,336,631 B2	12/2012	Shampine et al.	9,764,266 B1	9/2017	Carter
8,388,317 B2	3/2013	Sung	9,777,748 B2	10/2017	Lu et al.
8,414,673 B2	4/2013	Raje et al.	9,803,467 B2	10/2017	Tang et al.
8,469,826 B2	6/2013	Brosowske	9,803,793 B2	10/2017	Davi et al.
8,500,215 B2	8/2013	Gastauer	9,809,308 B2	11/2017	Aguilar et al.
8,506,267 B2	8/2013	Gambier et al.	9,829,002 B2	11/2017	Crom
8,575,873 B2	11/2013	Peterson et al.	9,840,897 B2	12/2017	Larson
8,616,005 B1	12/2013	Cousino, Sr. et al.	9,840,901 B2	12/2017	Oering et al.
8,621,873 B2	1/2014	Robertson et al.	9,845,730 B2	12/2017	Betti et al.
8,641,399 B2	2/2014	Mucibabic	9,850,422 B2	12/2017	Lestz et al.
8,656,990 B2	2/2014	Kajaria et al.	9,856,131 B1	1/2018	Moffitt
8,672,606 B2	3/2014	Glynn et al.	9,863,279 B2	1/2018	Laing et al.
8,707,853 B1	4/2014	Dille et al.	9,869,305 B1	1/2018	Crowe et al.
8,708,667 B2	4/2014	Collingborn	9,871,406 B1	1/2018	Churnock et al.
8,714,253 B2	5/2014	Sherwood et al.	9,879,609 B1	1/2018	Crowe et al.
8,757,918 B2	6/2014	Ramnarain et al.	RE46,725 E	2/2018	Case et al.
8,763,583 B2	7/2014	Hofbauer et al.	9,893,500 B2	2/2018	Oehring et al.
8,770,329 B2	7/2014	Spitler	9,893,660 B2	2/2018	Peterson et al.
8,784,081 B1	7/2014	Blume	9,897,003 B2	2/2018	Motakef et al.
8,789,601 B2	7/2014	Broussard et al.	9,920,615 B2	3/2018	Zhang et al.
8,794,307 B2	8/2014	Coquilleau et al.	9,945,365 B2	4/2018	Hernandez et al.
8,801,394 B2	8/2014	Anderson	9,964,052 B2	5/2018	Millican et al.
8,851,186 B2	10/2014	Shampine et al.	9,970,278 B2	5/2018	Broussard et al.
8,851,441 B2	10/2014	Acuna et al.	9,981,840 B2	5/2018	Shock
8,894,356 B2	11/2014	Lafontaine et al.	9,995,102 B2	6/2018	Dillie et al.
8,905,056 B2	12/2014	Kendrick	9,995,218 B2	6/2018	Oehring et al.
8,951,019 B2	2/2015	Hains et al.	10,008,880 B2	6/2018	Vicknair et al.
8,973,560 B2	3/2015	Krug	10,008,912 B2	6/2018	Davey et al.
8,997,904 B2	4/2015	Cryer et al.	10,018,096 B2	7/2018	Wallimann et al.
9,011,111 B2	4/2015	Lesko	10,020,711 B2	7/2018	Oehring et al.
9,016,383 B2	4/2015	Shampine et al.	10,024,123 B2	7/2018	Steffenhagen et al.
9,032,620 B2	5/2015	Frassinelli et al.	10,029,289 B2	7/2018	Wendorski et al.
9,057,247 B2	6/2015	Kumar et al.	10,030,579 B2	7/2018	Austin et al.
9,097,249 B2	8/2015	Petersen	10,036,238 B2	7/2018	Oehring
9,103,193 B2	8/2015	Coli et al.	10,040,541 B2	8/2018	Wilson et al.
9,121,257 B2	9/2015	Coli et al.	10,060,293 B2	8/2018	Del Bono
9,140,110 B2	9/2015	Coli et al.	10,060,349 B2	8/2018	Álvarez et al.
9,175,810 B2	11/2015	Hains	10,077,933 B2	9/2018	Nelson et al.
9,187,982 B2	11/2015	Dehring et al.	10,082,137 B2	9/2018	Graham et al.
			10,094,366 B2	10/2018	Marica
			10,100,827 B2	10/2018	Devan et al.
			10,107,084 B2	10/2018	Coli et al.
			10,107,085 B2	10/2018	Coli et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

10,114,061 B2	10/2018	Frampton et al.	10,961,914 B1	3/2021	Yeung et al.
10,119,381 B2	11/2018	Oehring et al.	10,961,993 B1	3/2021	Ji et al.
10,125,750 B2	11/2018	Pfaff	10,961,995 B2	3/2021	Mayorca
10,134,257 B2	11/2018	Zhang et al.	10,968,837 B1	4/2021	Yeung et al.
10,138,098 B2	11/2018	Sorensen et al.	10,982,523 B1	4/2021	Hill et al.
10,151,244 B2	12/2018	Giancotti et al.	10,989,019 B2	4/2021	Cai et al.
10,161,423 B2	12/2018	Rampen	10,989,180 B2	4/2021	Yeung et al.
10,174,599 B2	1/2019	Shampine et al.	10,995,564 B2	5/2021	Miller et al.
10,184,397 B2	1/2019	Austin et al.	11,002,189 B2	5/2021	Yeung et al.
10,196,258 B2	2/2019	Kalala et al.	11,008,950 B2	5/2021	Ethier et al.
10,221,856 B2	3/2019	Hernandez et al.	11,015,423 B1	5/2021	Yeung et al.
10,227,854 B2	3/2019	Glass	11,015,536 B2	5/2021	Yeung et al.
10,227,855 B2	3/2019	Coli et al.	11,015,594 B2	5/2021	Yeung et al.
10,246,984 B2	4/2019	Payne et al.	11,022,526 B1	6/2021	Yeung et al.
10,247,182 B2	4/2019	Zhang et al.	11,028,677 B1	6/2021	Yeung et al.
10,254,732 B2	4/2019	Oehring et al.	11,035,213 B2	6/2021	Dusterhoft et al.
10,267,439 B2	4/2019	Pryce et al.	11,035,214 B2	6/2021	Cui et al.
10,280,724 B2	5/2019	Hinderliter	11,047,379 B1	6/2021	Li et al.
10,287,943 B1	5/2019	Schiltz	11,053,853 B2	7/2021	Li et al.
10,288,519 B2	5/2019	De La Cruz	11,060,455 B1	7/2021	Yeung et al.
10,303,190 B2	5/2019	Shock	11,066,915 B1	7/2021	Yeung et al.
10,305,350 B2	5/2019	Johnson et al.	11,068,455 B2	7/2021	Shabi et al.
10,316,832 B2	6/2019	Byrne	11,085,281 B1	8/2021	Yeung et al.
10,317,875 B2	6/2019	Pandurangan et al.	11,085,282 B2	8/2021	Mazrooe et al.
10,337,402 B2	7/2019	Austin et al.	11,092,152 B2	8/2021	Yeung et al.
10,358,035 B2	7/2019	Cryer	11,098,651 B1	8/2021	Yeung et al.
10,371,012 B2	8/2019	Davis et al.	11,105,250 B1	8/2021	Zhang et al.
10,374,485 B2	8/2019	Morris et al.	11,105,266 B2	8/2021	Zhou et al.
10,378,326 B2	8/2019	Morris et al.	11,109,508 B1	8/2021	Yeung et al.
10,393,108 B2	8/2019	Chong et al.	11,111,768 B1	9/2021	Yeung et al.
10,407,990 B2	9/2019	Oehring et al.	11,125,066 B1	9/2021	Yeung et al.
10,408,031 B2	9/2019	Oehring et al.	11,125,156 B2	9/2021	Zhang et al.
10,415,348 B2	9/2019	Zhang et al.	11,129,295 B1	9/2021	Yeung et al.
10,415,557 B1	9/2019	Crowe et al.	11,143,000 B2	10/2021	Li et al.
10,415,562 B2	9/2019	Kajita et al.	11,143,005 B2	10/2021	Dusterhoft et al.
RE47,695 E	11/2019	Case et al.	11,143,006 B1	10/2021	Zhang et al.
10,465,689 B2	11/2019	Crom	11,149,533 B1	10/2021	Yeung et al.
10,478,753 B1	11/2019	Elms et al.	11,149,726 B1	10/2021	Yeung et al.
10,526,882 B2	1/2020	Oehring et al.	11,156,159 B1	10/2021	Yeung et al.
10,563,649 B2	2/2020	Zhang et al.	11,168,681 B2	11/2021	Boguski
10,577,910 B2	3/2020	Stephenson	11,174,716 B1	11/2021	Yeung et al.
10,584,645 B2	3/2020	Nakagawa et al.	11,193,360 B1	12/2021	Yeung et al.
10,590,867 B2	3/2020	Thomassin et al.	11,193,361 B1	12/2021	Yeung et al.
10,598,258 B2	3/2020	Oehring et al.	11,205,880 B1	12/2021	Yeung et al.
10,610,842 B2	4/2020	Chong	11,205,881 B2	12/2021	Yeung et al.
10,662,749 B1	5/2020	Hill et al.	11,208,879 B1	12/2021	Yeung et al.
10,711,787 B1	7/2020	Darley	11,208,953 B1	12/2021	Yeung et al.
10,738,580 B1	8/2020	Fischer et al.	11,220,895 B1	1/2022	Yeung et al.
10,753,153 B1	8/2020	Fischer et al.	11,236,739 B2	2/2022	Yeung et al.
10,753,165 B1	8/2020	Fischer et al.	11,242,737 B2	2/2022	Zhang et al.
10,760,556 B1	9/2020	Crom et al.	11,243,509 B2	2/2022	Cai et al.
10,794,165 B2	10/2020	Fischer et al.	11,251,650 B1	2/2022	Liu et al.
10,794,166 B2	10/2020	Reckels et al.	11,261,717 B2	3/2022	Yeung et al.
10,801,311 B1	10/2020	Cui et al.	11,268,346 B2	3/2022	Yeung et al.
10,815,764 B1	10/2020	Yeung et al.	11,280,266 B2	3/2022	Yeung et al.
10,815,978 B2	10/2020	Glass	11,306,835 B1	4/2022	Dille et al.
10,830,032 B1	11/2020	Zhang et al.	RE49,083 E	5/2022	Case et al.
10,830,225 B2	11/2020	Repaci	11,339,638 B1	5/2022	Yeung et al.
10,859,203 B1	12/2020	Cui et al.	11,346,200 B2	5/2022	Cai et al.
10,864,487 B1	12/2020	Han et al.	11,373,058 B2	6/2022	Jaaskelainen et al.
10,865,624 B1	12/2020	Cui et al.	RE49,140 E	7/2022	Case et al.
10,865,631 B1	12/2020	Zhang et al.	11,377,943 B2	7/2022	Kriebel et al.
10,870,093 B1	12/2020	Zhong et al.	RE49,155 E	8/2022	Case et al.
10,871,045 B2	12/2020	Fischer et al.	RE49,156 E	8/2022	Case et al.
10,892,596 B2	1/2021	Enya et al.	11,401,927 B2	8/2022	Li et al.
10,895,202 B1	1/2021	Yeung et al.	11,428,165 B2	8/2022	Yeung et al.
10,900,475 B2	1/2021	Weightman et al.	11,441,483 B2	9/2022	Li et al.
10,907,459 B1	2/2021	Yeung et al.	11,448,122 B2	9/2022	Feng et al.
10,927,774 B2	2/2021	Cai et al.	11,466,680 B2	10/2022	Yeung et al.
10,927,802 B2	2/2021	Oehring	11,480,040 B2	10/2022	Han et al.
10,954,770 B1	3/2021	Yeung et al.	11,492,887 B2	11/2022	Cui et al.
10,954,855 B1	3/2021	Ji et al.	11,499,405 B2	11/2022	Zhang et al.
10,961,614 B1	3/2021	Yeung et al.	11,506,039 B2	11/2022	Zhang et al.
10,961,908 B1	3/2021	Yeung et al.	11,512,570 B2	11/2022	Yeung
10,961,912 B1	3/2021	Yeung et al.	11,519,395 B2	12/2022	Zhang et al.
			11,519,405 B2	12/2022	Deng et al.
			11,530,602 B2	12/2022	Yeung et al.
			11,549,349 B2	1/2023	Wang et al.
			11,555,390 B2	1/2023	Cui et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

11,555,756 B2	1/2023	Yeung et al.	2010/0300683 A1	12/2010	Looper et al.
11,557,887 B2	1/2023	Ji et al.	2010/0310384 A1	12/2010	Stephenson et al.
11,560,779 B2	1/2023	Mao et al.	2011/0041681 A1	2/2011	Duerr
11,560,845 B2	1/2023	Yeung et al.	2011/0052423 A1	3/2011	Gambier et al.
11,572,775 B2	2/2023	Mao et al.	2011/0054704 A1	3/2011	Karpman et al.
11,575,249 B2	2/2023	Ji et al.	2011/0085924 A1	4/2011	Shampine et al.
11,592,020 B2	2/2023	Chang et al.	2011/0146244 A1	6/2011	Farman et al.
11,596,047 B2	2/2023	Liu et al.	2011/0146246 A1	6/2011	Farman et al.
11,598,263 B2	3/2023	Yeung et al.	2011/0173991 A1	7/2011	Dean
11,603,797 B2	3/2023	Zhang et al.	2011/0197988 A1	8/2011	Van Vliet et al.
11,607,982 B2	3/2023	Tian et al.	2011/0241888 A1	10/2011	Lu et al.
11,608,726 B2	3/2023	Zhang et al.	2011/0265443 A1	11/2011	Ansari
11,624,326 B2	4/2023	Yeung et al.	2011/0272158 A1	11/2011	Neal
11,629,583 B2	4/2023	Yeung et al.	2012/0023973 A1	2/2012	Mayorca
11,629,589 B2	4/2023	Lin et al.	2012/0048242 A1	3/2012	Surnilla et al.
11,649,766 B1	5/2023	Yeung et al.	2012/0085541 A1	4/2012	Love et al.
11,662,384 B2	5/2023	Liu et al.	2012/0137699 A1	6/2012	Montagne et al.
11,668,173 B2	6/2023	Zhang et al.	2012/0179444 A1	7/2012	Ganguly et al.
11,668,289 B2	6/2023	Chang et al.	2012/0192542 A1	8/2012	Chillar et al.
11,677,238 B2	6/2023	Liu et al.	2012/0199001 A1	8/2012	Chillar et al.
2002/0126922 A1	9/2002	Cheng et al.	2012/0204627 A1	8/2012	Anderl et al.
2002/0197176 A1	12/2002	Kondo	2012/0255734 A1	10/2012	Coli et al.
2003/0031568 A1	2/2003	Stiefel	2012/0310509 A1	12/2012	Pardo et al.
2003/0061819 A1	4/2003	Kuroki et al.	2012/0324903 A1	12/2012	Dewis et al.
2003/0161212 A1	8/2003	Neal et al.	2013/0068307 A1	3/2013	Hains et al.
2003/0218850 A1 *	11/2003	Kubo H01L 23/467 361/103	2013/0087045 A1	4/2013	Sullivan et al.
2004/0016245 A1	1/2004	Pierson	2013/0087945 A1	4/2013	Kusters et al.
2004/0074238 A1	4/2004	Wantanabe et al.	2013/0134702 A1	5/2013	Boraas et al.
2004/0076526 A1	4/2004	Fukano et al.	2013/0189915 A1	7/2013	Hazard
2004/0187950 A1	9/2004	Cohen et al.	2013/0205798 A1	8/2013	Kwok et al.
2004/0219040 A1	11/2004	Kugelev et al.	2013/0233165 A1	9/2013	Matzner et al.
2005/0051322 A1	3/2005	Speer	2013/0255953 A1	10/2013	Tudor
2005/0056081 A1	3/2005	Gocho	2013/0259707 A1	10/2013	Yin
2005/0139286 A1	6/2005	Poulter	2013/0284455 A1	10/2013	Kajaria et al.
2005/0196298 A1	9/2005	Manning	2013/0300341 A1	11/2013	Gillette
2005/0226754 A1	10/2005	Orr et al.	2013/0306322 A1	11/2013	Sanborn
2005/0274134 A1	12/2005	Ryu et al.	2014/0000668 A1	1/2014	Lessard
2006/0061091 A1	3/2006	Osterloh	2014/0010671 A1	1/2014	Cryer et al.
2006/0062914 A1	3/2006	Garg et al.	2014/0013768 A1	1/2014	Laing et al.
2006/0196251 A1	9/2006	Richy	2014/0032082 A1	1/2014	Gehrke et al.
2006/0211356 A1	9/2006	Grassman	2014/0044517 A1	2/2014	Saha et al.
2006/0228225 A1	10/2006	Rogers	2014/0048253 A1	2/2014	Andreychuk
2006/0260331 A1	11/2006	Andreychuk	2014/0090729 A1	4/2014	Coulter et al.
2006/0272333 A1	12/2006	Sundin	2014/0090742 A1	4/2014	Coskrey et al.
2007/0029090 A1	2/2007	Andreychuk et al.	2014/0094105 A1	4/2014	Lundh et al.
2007/0041848 A1	2/2007	Wood et al.	2014/0095114 A1	4/2014	Thomeer et al.
2007/0066406 A1	3/2007	Keller et al.	2014/0095554 A1	4/2014	Thomeer et al.
2007/0098580 A1	5/2007	Petersen	2014/0123621 A1	5/2014	Driessens et al.
2007/0107981 A1	5/2007	Sicotte	2014/0130422 A1	5/2014	Laing et al.
2007/0125544 A1	6/2007	Robinson et al.	2014/0138079 A1	5/2014	Broussard et al.
2007/0169543 A1	7/2007	Fazekas	2014/0144641 A1	5/2014	Chandler
2007/0181212 A1	8/2007	Fell	2014/0147291 A1	5/2014	Burnette
2007/0277982 A1	12/2007	Shampine et al.	2014/0158345 A1	6/2014	Jang et al.
2007/0295569 A1	12/2007	Manzoor et al.	2014/0174097 A1	6/2014	Hammer et al.
2008/0006089 A1	1/2008	Adnan et al.	2014/0196459 A1	7/2014	Futa et al.
2008/0098891 A1	5/2008	Feher	2014/0216736 A1	8/2014	Leugemors et al.
2008/0161974 A1	7/2008	Alston	2014/0219824 A1	8/2014	Burnette
2008/0212275 A1	9/2008	Waryck et al.	2014/0250845 A1	9/2014	Jackson et al.
2008/0229757 A1	9/2008	Alexander et al.	2014/0251623 A1	9/2014	Lestz et al.
2008/0264625 A1	10/2008	Ochoa	2014/0277772 A1	9/2014	Lopez et al.
2008/0264649 A1	10/2008	Crawford	2014/0290266 A1	10/2014	Veilleux, Jr. et al.
2008/0298982 A1	12/2008	Pabst	2014/0318638 A1	10/2014	Harwood et al.
2009/0064685 A1	3/2009	Busekros et al.	2014/0322050 A1	10/2014	Marette et al.
2009/0068031 A1	3/2009	Gambier et al.	2015/0027730 A1	1/2015	Hall et al.
2009/0092510 A1	4/2009	Williams et al.	2015/0078924 A1	3/2015	Zhang et al.
2009/0124191 A1	5/2009	Van Becelaere et al.	2015/0101344 A1	4/2015	Jarrier et al.
2009/0178412 A1	7/2009	Spytek	2015/0114652 A1	4/2015	Lestz et al.
2009/0212630 A1	8/2009	Fiegel et al.	2015/0129210 A1	5/2015	Chong et al.
2009/0249794 A1	10/2009	Wilkes et al.	2015/0135659 A1	5/2015	Jarrier et al.
2009/0252616 A1	10/2009	Brunet et al.	2015/0159553 A1	6/2015	Kippel et al.
2009/0308602 A1	12/2009	Bruins et al.	2015/0192117 A1	7/2015	Bridges
2010/0019626 A1	1/2010	Stout et al.	2015/0204148 A1	7/2015	Liu et al.
2010/0071899 A1	3/2010	Coquilleau et al.	2015/0204322 A1	7/2015	Iund et al.
2010/0218508 A1	9/2010	Brown et al.	2015/0211512 A1	7/2015	Wiegman et al.
			2015/0214816 A1	7/2015	Raad
			2015/0217672 A1	8/2015	Shampine et al.
			2015/0226140 A1	8/2015	Zhang et al.
			2015/0252661 A1	9/2015	Glass
			2015/0275891 A1	10/2015	Chong et al.

(56)	References Cited					
	U.S. PATENT DOCUMENTS					
2015/0337730	A1	11/2015	Kupiszewski et al.	2018/0034280	A1	2/2018 Pedersen
2015/0340864	A1	11/2015	Compton	2018/0038216	A1	2/2018 Zhang et al.
2015/0345385	A1	12/2015	Santini	2018/0038328	A1	2/2018 Louven et al.
2015/0369351	A1	12/2015	Hermann et al.	2018/0041093	A1	2/2018 Miranda
2016/0032703	A1	2/2016	Broussard et al.	2018/0045202	A1	2/2018 Crom
2016/0032836	A1	2/2016	Hawkinson et al.	2018/0058171	A1	3/2018 Roesner et al.
2016/0076447	A1	3/2016	Merlo et al.	2018/0087499	A1	3/2018 Zhang et al.
2016/0102581	A1	4/2016	Del Bono	2018/0087996	A1	3/2018 De La Cruz
2016/0105022	A1	4/2016	Oehring et al.	2018/0156210	A1	6/2018 Oehring et al.
2016/0108713	A1	4/2016	Dunaeva et al.	2018/0172294	A1	6/2018 Owen
2016/0123185	A1	5/2016	Le Pache et al.	2018/0183219	A1	6/2018 Oehring et al.
2016/0168979	A1	6/2016	Zhang et al.	2018/0186442	A1	7/2018 Maier
2016/0177675	A1	6/2016	Morris et al.	2018/0187662	A1	7/2018 Hill et al.
2016/0177945	A1	6/2016	Byrne et al.	2018/0209415	A1	7/2018 Zhang et al.
2016/0186671	A1	6/2016	Austin et al.	2018/0223640	A1	8/2018 Keihany et al.
2016/0195082	A1	7/2016	Wiegman et al.	2018/0224044	A1	8/2018 Penney
2016/0215774	A1	7/2016	Oklejas et al.	2018/0229998	A1	8/2018 Shock
2016/0230525	A1	8/2016	Lestz et al.	2018/0258746	A1	9/2018 Broussard et al.
2016/0244314	A1	8/2016	Van Vliet et al.	2018/0266412	A1	9/2018 Stokkevag et al.
2016/0248230	A1	8/2016	Tawy et al.	2018/0278124	A1	9/2018 Oehring et al.
2016/0253634	A1	9/2016	Thomeer et al.	2018/0283102	A1	10/2018 Cook
2016/0258267	A1	9/2016	Payne et al.	2018/0283618	A1	10/2018 Cook
2016/0273328	A1	9/2016	Oehring	2018/0284817	A1	10/2018 Cook et al.
2016/0273346	A1	9/2016	Tang et al.	2018/0290877	A1	10/2018 Shock
2016/0290114	A1	10/2016	Oehring et al.	2018/0291781	A1	10/2018 Pedrini
2016/0319650	A1	11/2016	Oehring et al.	2018/0298731	A1	10/2018 Bishop
2016/0326845	A1	11/2016	Djikpesse et al.	2018/0298735	A1	10/2018 Conrad
2016/0348479	A1	12/2016	Oehring et al.	2018/0307255	A1	10/2018 Bishop
2016/0369609	A1	12/2016	Morris et al.	2018/0313456	A1	11/2018 Bayyook et al.
2017/0009905	A1	1/2017	Arnold	2018/0328157	A1*	11/2018 Bishop F04B 17/06
2017/0016433	A1	1/2017	Chong et al.	2018/0334893	A1	11/2018 Oehring
2017/0030177	A1	2/2017	Oehring et al.	2018/0363435	A1	12/2018 Coli et al.
2017/0038137	A1	2/2017	Turney	2018/0363436	A1	12/2018 Coli et al.
2017/0045055	A1	2/2017	Hoefel et al.	2018/0363437	A1	12/2018 Coli et al.
2017/0052087	A1	2/2017	Faqihi et al.	2018/0363438	A1	12/2018 Coli et al.
2017/0074074	A1	3/2017	Joseph et al.	2019/0003272	A1	1/2019 Morris et al.
2017/0074076	A1	3/2017	Joseph et al.	2019/0003329	A1	1/2019 Morris et al.
2017/0074089	A1	3/2017	Agarwal et al.	2019/0009232	A1*	1/2019 Yeung B01F 35/2203
2017/0082110	A1	3/2017	Lammers	2019/0010793	A1	1/2019 Hinderliter
2017/0089189	A1	3/2017	Norris et al.	2019/0011051	A1	1/2019 Yeung
2017/0114613	A1	4/2017	Lecerf et al.	2019/0048993	A1	2/2019 Akiyama et al.
2017/0114625	A1	4/2017	Norris et al.	2019/0063263	A1	2/2019 Davis et al.
2017/0122310	A1	5/2017	Ladron de Guevara	2019/0063341	A1	2/2019 Davis
2017/0131174	A1	5/2017	Enev et al.	2019/0067991	A1	2/2019 Davis et al.
2017/0145918	A1	5/2017	Oehring et al.	2019/0071992	A1	3/2019 Feng
2017/0191350	A1	7/2017	Johns et al.	2019/0072005	A1	3/2019 Fisher et al.
2017/0218727	A1	8/2017	Oehring et al.	2019/0078471	A1	3/2019 Braglia et al.
2017/0226839	A1	8/2017	Broussard et al.	2019/0088845	A1	3/2019 Sugi et al.
2017/0226842	A1	8/2017	Omont et al.	2019/0091619	A1	3/2019 Huang
2017/0226998	A1	8/2017	Zhang et al.	2019/0106316	A1	4/2019 Van Vliet et al.
2017/0227002	A1	8/2017	Mikulski et al.	2019/0106970	A1	4/2019 Oehring
2017/0233103	A1	8/2017	Teicholz et al.	2019/0112908	A1	4/2019 Coli et al.
2017/0234165	A1*	8/2017	Kersey F02C 7/20 60/798	2019/0112910	A1	4/2019 Oehring et al.
2017/0234308	A1	8/2017	Buckley	2019/0119096	A1	4/2019 Haile et al.
2017/0241336	A1	8/2017	Jones et al.	2019/0120024	A1	4/2019 Oehring et al.
2017/0241671	A1	8/2017	Ahmad	2019/0120031	A1	4/2019 Gilje
2017/0248034	A1	8/2017	Dzieciol et al.	2019/0120134	A1	4/2019 Goleczka et al.
2017/0248208	A1	8/2017	Tamura	2019/0128247	A1	5/2019 Douglas, III
2017/0248308	A1	8/2017	Makarychev-Mikhailov et al.	2019/0128288	A1	5/2019 Konada et al.
2017/0275149	A1	9/2017	Schmidt	2019/0131607	A1	5/2019 Gillette
2017/0288400	A1	10/2017	Williams	2019/0136677	A1	5/2019 Shampine et al.
2017/0292409	A1	10/2017	Aguilar et al.	2019/0153843	A1	5/2019 Headrick
2017/0302135	A1	10/2017	Cory	2019/0153938	A1	5/2019 Hammoud
2017/0305736	A1	10/2017	Haile et al.	2019/0154020	A1	5/2019 Glass
2017/0306847	A1	10/2017	Suciu et al.	2019/0155318	A1	5/2019 Meunier
2017/0306936	A1	10/2017	Dole	2019/0178234	A1	6/2019 Beisel
2017/0322086	A1	11/2017	Luharuka	2019/0178235	A1	6/2019 Coskrey et al.
2017/0333086	A1	11/2017	Jackson	2019/0185312	A1	6/2019 Bush et al.
2017/0334448	A1	11/2017	Schwunk	2019/0203572	A1	7/2019 Morris et al.
2017/0335842	A1	11/2017	Robinson et al.	2019/0204021	A1	7/2019 Morris et al.
2017/0350471	A1	12/2017	Steidl et al.	2019/0211661	A1	7/2019 Reckles et al.
2017/0356470	A1	12/2017	Jaffrey	2019/0211814	A1	7/2019 Weightman et al.
2017/0370199	A1	12/2017	Witkowski et al.	2019/0217258	A1	7/2019 Bishop
2017/0370480	A1	12/2017	Witkowski et al.	2019/0226317	A1	7/2019 Payne et al.
				2019/0245348	A1	8/2019 Hinderliter et al.
				2019/0249652	A1	8/2019 Stephenson et al.
				2019/0249754	A1	8/2019 Oehring et al.
				2019/0257297	A1	8/2019 Botting et al.
				2019/0264667	A1	8/2019 Byrne

(56)

References Cited

U.S. PATENT DOCUMENTS

2019/0277279	A1	9/2019	Byrne et al.	2020/0392827	A1	12/2020	George et al.
2019/0277295	A1	9/2019	Clyburn et al.	2020/0393088	A1	12/2020	Sizemore et al.
2019/0309585	A1	10/2019	Miller et al.	2020/0398238	A1	12/2020	Zhong et al.
2019/0316447	A1	10/2019	Oehring et al.	2020/0400000	A1	12/2020	Ghasripoor et al.
2019/0316456	A1	10/2019	Beisel et al.	2020/0400005	A1	12/2020	Han et al.
2019/0323337	A1	10/2019	Glass et al.	2020/0407625	A1	12/2020	Stephenson
2019/0330923	A1	10/2019	Gable et al.	2020/0408071	A1	12/2020	Li et al.
2019/0331117	A1	10/2019	Gable et al.	2020/0408144	A1	12/2020	Feng et al.
2019/0337392	A1	11/2019	Joshi et al.	2020/0408147	A1	12/2020	Zhang et al.
2019/0338762	A1	11/2019	Curry et al.	2020/0408149	A1	12/2020	Li et al.
2019/0345920	A1	11/2019	Surjaatmadja et al.	2021/0025324	A1	1/2021	Morris et al.
2019/0353103	A1	11/2019	Roberge	2021/0025383	A1	1/2021	Bodishbaugh et al.
2019/0356199	A1	11/2019	Morris et al.	2021/0032961	A1	2/2021	Hinderliter et al.
2019/0376449	A1	12/2019	Carrell	2021/0054727	A1	2/2021	Floyd
2019/0383123	A1	12/2019	Hinderliter	2021/0071503	A1	3/2021	Ogg et al.
2020/0003205	A1	1/2020	Stokkevåg et al.	2021/0071574	A1	3/2021	Feng et al.
2020/0011165	A1	1/2020	George et al.	2021/0071579	A1	3/2021	Li et al.
2020/0025058	A1*	1/2020	Jochman F01P 11/10	2021/0071654	A1	3/2021	Brunson
2020/0040878	A1	2/2020	Morris	2021/0071752	A1	3/2021	Cui et al.
2020/0049136	A1	2/2020	Stephenson	2021/0079758	A1	3/2021	Yeung et al.
2020/0049153	A1	2/2020	Headrick et al.	2021/0079851	A1	3/2021	Yeung et al.
2020/0071998	A1	3/2020	Oehring et al.	2021/0086851	A1	3/2021	Zhang et al.
2020/0072201	A1	3/2020	Marica	2021/0087883	A1	3/2021	Zhang et al.
2020/0088202	A1	3/2020	Sigmar et al.	2021/0087916	A1	3/2021	Zhang et al.
2020/0095854	A1	3/2020	Hinderliter	2021/0087925	A1	3/2021	Heidari et al.
2020/0109610	A1	4/2020	Husoy et al.	2021/0087943	A1	3/2021	Cui et al.
2020/0109616	A1	4/2020	Oehring et al.	2021/0088042	A1	3/2021	Zhang et al.
2020/0123729	A1*	4/2020	Guender E02D 27/44	2021/0123425	A1	4/2021	Cui et al.
2020/0132058	A1	4/2020	Mollatt	2021/0123434	A1	4/2021	Cui et al.
2020/0141219	A1	5/2020	Oehring et al.	2021/0123435	A1	4/2021	Cui et al.
2020/0141326	A1	5/2020	Redford et al.	2021/0131409	A1	5/2021	Cui et al.
2020/0141907	A1	5/2020	Meck et al.	2021/0140416	A1	5/2021	Buckley
2020/0166026	A1	5/2020	Marica	2021/0148208	A1	5/2021	Thomas et al.
2020/0206704	A1	7/2020	Chong	2021/0156240	A1	5/2021	Cicci et al.
2020/0208733	A1	7/2020	Kim	2021/0156241	A1	5/2021	Cook
2020/0223648	A1	7/2020	Herman et al.	2021/0172282	A1	6/2021	Wang et al.
2020/0224645	A1	7/2020	Buckley	2021/0180517	A1	6/2021	Zhou et al.
2020/0232454	A1	7/2020	Chretien et al.	2021/0190045	A1	6/2021	Zhang et al.
2020/0256333	A1	8/2020	Surjaatmadja	2021/0199110	A1	7/2021	Albert et al.
2020/0263498	A1	8/2020	Fischer et al.	2021/0222690	A1	7/2021	Beisel
2020/0263525	A1	8/2020	Reid	2021/0239112	A1	8/2021	Buckley
2020/0263526	A1	8/2020	Fischer et al.	2021/0246774	A1	8/2021	Cui et al.
2020/0263527	A1	8/2020	Fischer et al.	2021/0270261	A1	9/2021	Zhang et al.
2020/0263528	A1	8/2020	Fischer et al.	2021/0270264	A1	9/2021	Byrne
2020/0267888	A1	8/2020	Putz	2021/0285311	A1	9/2021	Ji et al.
2020/0291731	A1	9/2020	Haiderer et al.	2021/0285432	A1	9/2021	Ji et al.
2020/0295574	A1	9/2020	Batsch-Smith	2021/0301807	A1	9/2021	Cui et al.
2020/0300050	A1	9/2020	Oehring et al.	2021/0306720	A1	9/2021	Sandoval et al.
2020/0309027	A1	10/2020	Ryttonen	2021/0308638	A1	10/2021	Zhong et al.
2020/0309113	A1	10/2020	Hunter et al.	2021/0324718	A1	10/2021	Anders
2020/0325752	A1	10/2020	Clark et al.	2021/0348475	A1	11/2021	Yeung et al.
2020/0325760	A1	10/2020	Markham	2021/0348476	A1	11/2021	Yeung et al.
2020/0325761	A1	10/2020	Williams	2021/0348477	A1	11/2021	Yeung et al.
2020/0325791	A1	10/2020	Himmelman	2021/0355927	A1	11/2021	Jian et al.
2020/0325893	A1	10/2020	Kraige et al.	2021/0372394	A1	12/2021	Bagulayan et al.
2020/0332784	A1	10/2020	Zhang et al.	2021/0372395	A1	12/2021	Li et al.
2020/0332788	A1	10/2020	Cui et al.	2021/0376413	A1	12/2021	Asfha
2020/0340313	A1	10/2020	Fischer et al.	2021/0388760	A1	12/2021	Feng et al.
2020/0340322	A1	10/2020	Sizemore et al.	2022/0082007	A1	3/2022	Zhang et al.
2020/0340340	A1	10/2020	Oehring et al.	2022/0090476	A1	3/2022	Zhang et al.
2020/0340344	A1	10/2020	Reckels et al.	2022/0090477	A1	3/2022	Zhang et al.
2020/0340404	A1	10/2020	Stockstill	2022/0090478	A1	3/2022	Zhang et al.
2020/0347725	A1	11/2020	Morris et al.	2022/0112892	A1	4/2022	Cui et al.
2020/0354928	A1	11/2020	Wehler et al.	2022/0120262	A1	4/2022	Ji et al.
2020/0355055	A1	11/2020	Dusterhoft et al.	2022/0145740	A1	5/2022	Yuan et al.
2020/0362760	A1	11/2020	Morenko et al.	2022/0154775	A1	5/2022	Liu et al.
2020/0362764	A1	11/2020	Saintignan et al.	2022/0155373	A1	5/2022	Liu et al.
2020/0370394	A1	11/2020	Cai et al.	2022/0162931	A1	5/2022	Zhong et al.
2020/0370408	A1	11/2020	Cai et al.	2022/0162991	A1	5/2022	Zhang et al.
2020/0370429	A1	11/2020	Cai et al.	2022/0181859	A1	6/2022	Ji et al.
2020/0371490	A1	11/2020	Cai et al.	2022/0186724	A1	6/2022	Chang et al.
2020/0386169	A1	12/2020	Hinderliter et al.	2022/0213777	A1	7/2022	Cui et al.
2020/0386222	A1	12/2020	Pham et al.	2022/0220836	A1	7/2022	Zhang et al.
2020/0388140	A1	12/2020	Gomez et al.	2022/0224087	A1	7/2022	Ji et al.
2020/0392826	A1	12/2020	Cui et al.	2022/0228468	A1	7/2022	Cui et al.
				2022/0228469	A1	7/2022	Zhang et al.
				2022/0235639	A1	7/2022	Zhang et al.
				2022/0235640	A1	7/2022	Mao et al.
				2022/0235641	A1	7/2022	Zhang et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

2022/0235642 A1 7/2022 Zhang et al.
 2022/0235802 A1 7/2022 Jiang et al.
 2022/0242297 A1 8/2022 Tian et al.
 2022/0243613 A1 8/2022 Ji et al.
 2022/0243724 A1 8/2022 Li et al.
 2022/0250000 A1 8/2022 Zhang et al.
 2022/0255319 A1 8/2022 Liu et al.
 2022/0258659 A1 8/2022 Cui et al.
 2022/0259947 A1 8/2022 Li et al.
 2022/0259964 A1 8/2022 Zhang et al.
 2022/0268201 A1 8/2022 Feng et al.
 2022/0282606 A1 9/2022 Zhong et al.
 2022/0282726 A1 9/2022 Zhang et al.
 2022/0290549 A1 9/2022 Zhang et al.
 2022/0294194 A1 9/2022 Cao et al.
 2022/0298906 A1 9/2022 Zhong et al.
 2022/0307359 A1 9/2022 Liu et al.
 2022/0307424 A1 9/2022 Wang et al.
 2022/0314248 A1 10/2022 Ge et al.
 2022/0315347 A1 10/2022 Liu et al.
 2022/0316306 A1 10/2022 Liu et al.
 2022/0316362 A1 10/2022 Zhang et al.
 2022/0316461 A1 10/2022 Wang et al.
 2022/0325608 A1 10/2022 Zhang et al.
 2022/0330411 A1 10/2022 Liu et al.
 2022/0333471 A1 10/2022 Zhong et al.
 2022/0339646 A1 10/2022 Yu et al.
 2022/0341358 A1 10/2022 Ji et al.
 2022/0341362 A1 10/2022 Feng et al.
 2022/0341415 A1 10/2022 Deng et al.
 2022/0345007 A1 10/2022 Liu et al.
 2022/0349345 A1 11/2022 Zhang et al.
 2022/0353980 A1 11/2022 Liu et al.
 2022/0361309 A1 11/2022 Liu et al.
 2022/0364452 A1 11/2022 Wang et al.
 2022/0364453 A1 11/2022 Chang et al.
 2022/0372865 A1 11/2022 Lin et al.
 2022/0376280 A1 11/2022 Shao et al.
 2022/0381126 A1 12/2022 Cui et al.
 2022/0389799 A1 12/2022 Mao
 2022/0389803 A1 12/2022 Zhang et al.
 2022/0389804 A1 12/2022 Cui et al.
 2022/0389865 A1 12/2022 Feng et al.
 2022/0389867 A1 12/2022 Li et al.
 2022/0412196 A1 12/2022 Cui et al.
 2022/0412199 A1 12/2022 Mao et al.
 2022/0412200 A1 12/2022 Zhang et al.
 2022/0412258 A1 12/2022 Li et al.
 2022/0412379 A1 12/2022 Wang et al.
 2023/0001524 A1 1/2023 Jiang et al.
 2023/0003238 A1 1/2023 Du et al.
 2023/0015132 A1 1/2023 Feng et al.
 2023/0015529 A1 1/2023 Zhang et al.
 2023/0015581 A1 1/2023 Ji et al.
 2023/0017968 A1 1/2023 Deng et al.
 2023/0029574 A1 2/2023 Zhang et al.
 2023/0029671 A1 2/2023 Han et al.
 2023/0036118 A1 2/2023 Xing et al.
 2023/0040970 A1 2/2023 Liu et al.
 2023/0042379 A1 2/2023 Zhang et al.
 2023/0047033 A1 2/2023 Fu et al.
 2023/0048551 A1 2/2023 Feng et al.
 2023/0049462 A1 2/2023 Zhang et al.
 2023/0064964 A1 3/2023 Wang et al.
 2023/0074794 A1 3/2023 Liu et al.
 2023/0085124 A1 3/2023 Zhong et al.
 2023/0092506 A1 3/2023 Zhong et al.
 2023/0092705 A1 3/2023 Liu et al.
 2023/0106683 A1 4/2023 Zhang et al.
 2023/0107300 A1 4/2023 Huang et al.
 2023/0107791 A1 4/2023 Zhang et al.
 2023/0109018 A1 4/2023 Du et al.
 2023/0116458 A1 4/2023 Liu et al.
 2023/0117362 A1 4/2023 Zhang et al.
 2023/0119725 A1 4/2023 Wang et al.

2023/0119876 A1 4/2023 Mao et al.
 2023/0119896 A1 4/2023 Zhang et al.
 2023/0120810 A1 4/2023 Fu et al.
 2023/0121251 A1 4/2023 Cui et al.
 2023/0124444 A1 4/2023 Chang et al.
 2023/0138582 A1 5/2023 Li et al.
 2023/0144116 A1 5/2023 Li et al.
 2023/0145963 A1 5/2023 Zhang et al.
 2023/0151722 A1 5/2023 Cui et al.
 2023/0151723 A1 5/2023 Ji et al.
 2023/0152793 A1 5/2023 Wang et al.
 2023/0160289 A1 5/2023 Cui et al.
 2023/0160510 A1 5/2023 Bao et al.
 2023/0163580 A1 5/2023 Ji et al.
 2023/0167776 A1 6/2023 Cui et al.

FOREIGN PATENT DOCUMENTS

CA 2043184 8/1994
 CA 2829762 9/2012
 CA 2737321 9/2013
 CA 2876687 A1 5/2014
 CA 2693567 9/2014
 CA 2964597 10/2017
 CA 2876687 C 4/2019
 CA 3138533 11/2020
 CA 2919175 3/2021
 CN 2622404 6/2004
 CN 2779054 5/2006
 CN 2890325 4/2007
 CN 200964929 Y 10/2007
 CN 101323151 A 12/2008
 CN 201190660 Y 2/2009
 CN 201190892 Y 2/2009
 CN 201190893 Y 2/2009
 CN 101414171 A 4/2009
 CN 201215073 Y 4/2009
 CN 201236650 Y 5/2009
 CN 201275542 Y 7/2009
 CN 201275801 Y 7/2009
 CN 201333385 Y 10/2009
 CN 201443300 U 4/2010
 CN 201496415 U 6/2010
 CN 201501365 U 6/2010
 CN 201507271 U 6/2010
 CN 101323151 B 7/2010
 CN 201560210 U 8/2010
 CN 201581862 U 9/2010
 CN 201610728 U 10/2010
 CN 201610751 U 10/2010
 CN 201618530 U 11/2010
 CN 201661255 U 12/2010
 CN 101949382 1/2011
 CN 201756927 U 3/2011
 CN 101414171 B 5/2011
 CN 102128011 A 7/2011
 CN 102140898 A 8/2011
 CN 102155172 A 8/2011
 CN 102182904 9/2011
 CN 202000930 U 10/2011
 CN 202055781 U 11/2011
 CN 202082265 U 12/2011
 CN 202100216 U 1/2012
 CN 202100217 U 1/2012
 CN 202100815 U 1/2012
 CN 202124340 U 1/2012
 CN 202140051 U 2/2012
 CN 202140080 U 2/2012
 CN 202144789 U 2/2012
 CN 202144943 U 2/2012
 CN 202149354 U 2/2012
 CN 102383748 A 3/2012
 CN 202156297 U 3/2012
 CN 202158355 U 3/2012
 CN 202163504 U 3/2012
 CN 202165236 U 3/2012
 CN 202180866 U 4/2012
 CN 202181875 U 4/2012
 CN 202187744 U 4/2012

(56)	References Cited			CN	203531883	U	4/2014
	FOREIGN PATENT DOCUMENTS			CN	203556164	U	4/2014
				CN	203558809	U	4/2014
				CN	203559861	U	4/2014
				CN	203559893	U	4/2014
CN	202191854	U	4/2012	CN	203560189	U	4/2014
CN	202250008	U	5/2012	CN	102704870	B	5/2014
CN	101885307		7/2012	CN	203611843	U	5/2014
CN	102562020	A	7/2012	CN	203612531	U	5/2014
CN	202326156	U	7/2012	CN	203612843	U	5/2014
CN	202370773	U	8/2012	CN	203614062	U	5/2014
CN	202417397	U	9/2012	CN	203614388	U	5/2014
CN	202417461	U	9/2012	CN	203621045	U	6/2014
CN	102729335	A	10/2012	CN	203621046	U	6/2014
CN	202463955	U	10/2012	CN	203621051	U	6/2014
CN	202463957	U	10/2012	CN	203640993	U	6/2014
CN	202467739	U	10/2012	CN	203655221	U	6/2014
CN	202467801	U	10/2012	CN	103899280	A	7/2014
CN	202531016	U	11/2012	CN	103923670	A	7/2014
CN	202544794	U	11/2012	CN	203685052	U	7/2014
CN	102825039	A	12/2012	CN	203716936	U	7/2014
CN	202578592	U	12/2012	CN	103990410	A	8/2014
CN	202579164	U	12/2012	CN	103993869	A	8/2014
CN	202594808	U	12/2012	CN	203754009	U	8/2014
CN	202594928	U	12/2012	CN	203754025	U	8/2014
CN	202596615	U	12/2012	CN	203754341	U	8/2014
CN	202596616	U	12/2012	CN	203756614	U	8/2014
CN	102849880	A	1/2013	CN	203770264	U	8/2014
CN	102889191	A	1/2013	CN	203784519	U	8/2014
CN	202641535	U	1/2013	CN	203784520	U	8/2014
CN	202645475	U	1/2013	CN	104057864	A	9/2014
CN	202666716	U	1/2013	CN	203819819	U	9/2014
CN	202669645	U	1/2013	CN	203823431	U	9/2014
CN	202669944	U	1/2013	CN	203835337	U	9/2014
CN	202671336	U	1/2013	CN	104074500	A	10/2014
CN	202673269	U	1/2013	CN	203876633	U	10/2014
CN	202751982	U	2/2013	CN	203876636	U	10/2014
CN	102963629	A	3/2013	CN	203877364	U	10/2014
CN	202767964	U	3/2013	CN	203877365	U	10/2014
CN	202789791	U	3/2013	CN	203877375	U	10/2014
CN	202789792	U	3/2013	CN	203877424	U	10/2014
CN	202810717	U	3/2013	CN	203879476	U	10/2014
CN	202827276	U	3/2013	CN	203879479	U	10/2014
CN	202833093	U	3/2013	CN	203890292	U	10/2014
CN	202833370	U	3/2013	CN	203899476	U	10/2014
CN	102140898	B	4/2013	CN	203906206	U	10/2014
CN	202895467	U	4/2013	CN	104150728	A	11/2014
CN	202926404	U	5/2013	CN	104176522	A	12/2014
CN	202935216	U	5/2013	CN	104196464	A	12/2014
CN	202935798	U	5/2013	CN	104234651	A	12/2014
CN	202935816	U	5/2013	CN	203971841	U	12/2014
CN	202970631	U	6/2013	CN	203975450	U	12/2014
CN	103223315	A	7/2013	CN	204020788	U	12/2014
CN	203050598	U	7/2013	CN	204021980	U	12/2014
CN	103233714	A	8/2013	CN	204024625	U	12/2014
CN	103233715	A	8/2013	CN	204051401	U	12/2014
CN	103245523	A	8/2013	CN	204060661	U	12/2014
CN	103247220	A	8/2013	CN	104260672	A	1/2015
CN	103253839	A	8/2013	CN	104314512	A	1/2015
CN	103277290	A	9/2013	CN	204077478	U	1/2015
CN	103321782	A	9/2013	CN	204077526	U	1/2015
CN	203170270	U	9/2013	CN	204078307	U	1/2015
CN	203172509	U	9/2013	CN	204083051	U	1/2015
CN	203175778	U	9/2013	CN	204113168	U	1/2015
CN	203175787	U	9/2013	CN	104340682	A	2/2015
CN	102849880	B	10/2013	CN	104358536	A	2/2015
CN	203241231	U	10/2013	CN	104369687	A	2/2015
CN	203244941	U	10/2013	CN	104402178	A	3/2015
CN	203244942	U	10/2013	CN	104402185	A	3/2015
CN	203303798	U	11/2013	CN	104402186	A	3/2015
CN	102155172	B	12/2013	CN	204209819	U	3/2015
CN	102729335	B	12/2013	CN	204224560	U	3/2015
CN	103420532	A	12/2013	CN	204225813	U	3/2015
CN	203321792	U	12/2013	CN	204225839	U	3/2015
CN	203412658		1/2014	CN	104533392	A	4/2015
CN	203420697	U	2/2014	CN	104563938	A	4/2015
CN	203480755	U	3/2014	CN	104563994	A	4/2015
CN	103711437	A	4/2014	CN	104563995	A	4/2015
CN	203531815	U	4/2014	CN	104563998	A	4/2015
CN	203531871	U	4/2014				

(56)	References Cited			CN	106246120	A	12/2016
	FOREIGN PATENT DOCUMENTS			CN	205805471		12/2016
				CN	106321045	A	1/2017
				CN	205858306		1/2017
CN	104564033	A	4/2015	CN	106438310	A	2/2017
CN	204257122	U	4/2015	CN	205937833		2/2017
CN	204283610	U	4/2015	CN	104563994	B	3/2017
CN	204283782	U	4/2015	CN	206129196		4/2017
CN	204297682	U	4/2015	CN	104369687	B	5/2017
CN	204299810	U	4/2015	CN	106715165		5/2017
CN	103223315	B	5/2015	CN	106761561	A	5/2017
CN	104594857	A	5/2015	CN	105240064	B	6/2017
CN	104595493	A	5/2015	CN	206237147		6/2017
CN	104612647	A	5/2015	CN	206287832		6/2017
CN	104612928	A	5/2015	CN	206346711		7/2017
CN	104632126	A	5/2015	CN	104563995	B	9/2017
CN	204325094	U	5/2015	CN	107120822		9/2017
CN	204325098	U	5/2015	CN	107143298	A	9/2017
CN	204326983	U	5/2015	CN	107159046	A	9/2017
CN	204326985	U	5/2015	CN	107188018	A	9/2017
CN	204344040	U	5/2015	CN	206496016		9/2017
CN	204344095	U	5/2015	CN	104564033	B	10/2017
CN	104727797	A	6/2015	CN	107234358	A	10/2017
CN	204402414	U	6/2015	CN	107261975	A	10/2017
CN	204402423	U	6/2015	CN	206581929		10/2017
CN	204402450	U	6/2015	CN	104820372	B	12/2017
CN	103247220	B	7/2015	CN	105092401	B	12/2017
CN	104803568	A	7/2015	CN	107476769	A	12/2017
CN	204436360	U	7/2015	CN	107520526	A	12/2017
CN	204457524	U	7/2015	CN	206754664		12/2017
CN	204472485	U	7/2015	CN	107605427	A	1/2018
CN	204473625	U	7/2015	CN	106438310	B	2/2018
CN	204477303	U	7/2015	CN	107654196	A	2/2018
CN	204493095	U	7/2015	CN	107656499	A	2/2018
CN	204493309	U	7/2015	CN	107728657	A	2/2018
CN	103253839	B	8/2015	CN	206985503		2/2018
CN	104820372	A	8/2015	CN	207017968		2/2018
CN	104832093	A	8/2015	CN	107859053	A	3/2018
CN	104863523	A	8/2015	CN	207057867		3/2018
CN	204552723	U	8/2015	CN	207085817		3/2018
CN	204553866	U	8/2015	CN	105545207	B	4/2018
CN	204571831	U	8/2015	CN	107883091	A	4/2018
CN	204703814	U	10/2015	CN	107902427	A	4/2018
CN	204703833	U	10/2015	CN	107939290	A	4/2018
CN	204703834	U	10/2015	CN	107956708		4/2018
CN	105092401	A	11/2015	CN	207169595		4/2018
CN	103233715	B	12/2015	CN	207194873		4/2018
CN	103790927		12/2015	CN	207245674		4/2018
CN	105207097		12/2015	CN	108034466	A	5/2018
CN	204831952	U	12/2015	CN	108036071	A	5/2018
CN	204899777	U	12/2015	CN	108087050	A	5/2018
CN	102602323		1/2016	CN	207380566		5/2018
CN	105240064	A	1/2016	CN	108103483	A	6/2018
CN	204944834		1/2016	CN	108179046	A	6/2018
CN	205042127	U	2/2016	CN	108254276	A	7/2018
CN	205172478	U	4/2016	CN	108311535	A	7/2018
CN	103993869	B	5/2016	CN	207583576		7/2018
CN	105536299	A	5/2016	CN	207634064		7/2018
CN	105545207	A	5/2016	CN	207648054		7/2018
CN	205260249		5/2016	CN	207650621		7/2018
CN	103233714	B	6/2016	CN	108371894	A	8/2018
CN	104340682	B	6/2016	CN	207777153		8/2018
CN	205297518	U	6/2016	CN	108547601	A	9/2018
CN	205298447	U	6/2016	CN	108547766	A	9/2018
CN	205391821	U	7/2016	CN	108555826	A	9/2018
CN	205400701	U	7/2016	CN	108561098	A	9/2018
CN	103277290	B	8/2016	CN	108561750	A	9/2018
CN	104260672	B	8/2016	CN	108590617	A	9/2018
CN	205477370	U	8/2016	CN	207813495		9/2018
CN	205479153	U	8/2016	CN	207814698		9/2018
CN	205503058	U	8/2016	CN	207862275		9/2018
CN	205503068	U	8/2016	CN	108687954	A	10/2018
CN	205503089	U	8/2016	CN	207935270		10/2018
CN	105958098	A	9/2016	CN	207961582		10/2018
CN	205599180		9/2016	CN	207964530		10/2018
CN	205599180	U	9/2016	CN	108789848	A	11/2018
CN	106121577	A	11/2016	CN	108799473		11/2018
CN	205709587		11/2016	CN	108868675	A	11/2018
CN	104612928	B	12/2016	CN	208086829		11/2018

(56)	References Cited		CN	110485983 A	11/2019
	FOREIGN PATENT DOCUMENTS		CN	110485984 A	11/2019
			CN	110486249 A	11/2019
			CN	110500255 A	11/2019
CN	208089263	11/2018	CN	110510771 A	11/2019
CN	208169068	11/2018	CN	110513097 A	11/2019
CN	108979569 A	12/2018	CN	209650738	11/2019
CN	109027662 A	12/2018	CN	209653968	11/2019
CN	109058092 A	12/2018	CN	209654004	11/2019
CN	208179454	12/2018	CN	209654022	11/2019
CN	208179502	12/2018	CN	209654128	11/2019
CN	208253147	12/2018	CN	209656622	11/2019
CN	208260574	12/2018	CN	107849130 B	12/2019
CN	109114418 A	1/2019	CN	108087050 B	12/2019
CN	109141990 A	1/2019	CN	110566173 A	12/2019
CN	208313120	1/2019	CN	110608030 A	12/2019
CN	208330319	1/2019	CN	110617187 A	12/2019
CN	208342730	1/2019	CN	110617188 A	12/2019
CN	208430982	1/2019	CN	110617318 A	12/2019
CN	208430986	1/2019	CN	209740823	12/2019
CN	109404274 A	3/2019	CN	209780827	12/2019
CN	109429610 A	3/2019	CN	209798631	12/2019
CN	109491318 A	3/2019	CN	209799942	12/2019
CN	109515177 A	3/2019	CN	209800178	12/2019
CN	109526523 A	3/2019	CN	209855723	12/2019
CN	109534737 A	3/2019	CN	209855742	12/2019
CN	208564504	3/2019	CN	209875063	12/2019
CN	208564516	3/2019	CN	110656919 A	1/2020
CN	208564525	3/2019	CN	107520526 B	2/2020
CN	208564918	3/2019	CN	110787667 A	2/2020
CN	208576026	3/2019	CN	110821464 A	2/2020
CN	208576042	3/2019	CN	110833665 A	2/2020
CN	208650818	3/2019	CN	110848028 A	2/2020
CN	208669244	3/2019	CN	210049880	2/2020
CN	109555484 A	4/2019	CN	210049882	2/2020
CN	109682881 A	4/2019	CN	210097596	2/2020
CN	208730959	4/2019	CN	210105817	2/2020
CN	208735264	4/2019	CN	210105818	2/2020
CN	208746733	4/2019	CN	210105993	2/2020
CN	208749529	4/2019	CN	110873093 A	3/2020
CN	208750405	4/2019	CN	210139911	3/2020
CN	208764658	4/2019	CN	110947681 A	4/2020
CN	109736740 A	5/2019	CN	111058810 A	4/2020
CN	109751007 A	5/2019	CN	111075391 A	4/2020
CN	208868428	5/2019	CN	210289931	4/2020
CN	208870761	5/2019	CN	210289932	4/2020
CN	109869294 A	6/2019	CN	210289933	4/2020
CN	109882144 A	6/2019	CN	210303516	4/2020
CN	109882372 A	6/2019	CN	211412945	4/2020
CN	209012047	6/2019	CN	111089003 A	5/2020
CN	209100025	7/2019	CN	111151186 A	5/2020
CN	110080707 A	8/2019	CN	111167769 A	5/2020
CN	110118127 A	8/2019	CN	111169833 A	5/2020
CN	110124574 A	8/2019	CN	111173476 A	5/2020
CN	110145277 A	8/2019	CN	111185460 A	5/2020
CN	110145399 A	8/2019	CN	111185461 A	5/2020
CN	110152552 A	8/2019	CN	111188763 A	5/2020
CN	110155193 A	8/2019	CN	111206901 A	5/2020
CN	110159225 A	8/2019	CN	111206992 A	5/2020
CN	110159432	8/2019	CN	111206994 A	5/2020
CN	110159432 A	8/2019	CN	210449044	5/2020
CN	110159433 A	8/2019	CN	210460875	5/2020
CN	110208100 A	9/2019	CN	210522432	5/2020
CN	110252191 A	9/2019	CN	210598943	5/2020
CN	110284854 A	9/2019	CN	210598945	5/2020
CN	110284972 A	9/2019	CN	210598946	5/2020
CN	209387358	9/2019	CN	210599194	5/2020
CN	110374745 A	10/2019	CN	210599303	5/2020
CN	209534736	10/2019	CN	210600110	5/2020
CN	110425105 A	11/2019	CN	111219326 A	6/2020
CN	110439779 A	11/2019	CN	111350595 A	6/2020
CN	110454285 A	11/2019	CN	210660319	6/2020
CN	110454352 A	11/2019	CN	210714569	6/2020
CN	110467298 A	11/2019	CN	210769168	6/2020
CN	110469312 A	11/2019	CN	210769169	6/2020
CN	110469314 A	11/2019	CN	210769170	6/2020
CN	110469405 A	11/2019	CN	210770133	6/2020
CN	110469654 A	11/2019	CN	210825844	6/2020
CN	110485982 A	11/2019	CN	210888904	6/2020

(56)

References Cited

FOREIGN PATENT DOCUMENTS

CN	210888905	6/2020
CN	210889242	6/2020
CN	111397474 A	7/2020
CN	111412064 A	7/2020
CN	111441923 A	7/2020
CN	111441925 A	7/2020
CN	111503517 A	8/2020
CN	111515898 A	8/2020
CN	111594059 A	8/2020
CN	111594062 A	8/2020
CN	111594144 A	8/2020
CN	211201919	8/2020
CN	211201920	8/2020
CN	211202218	8/2020
CN	111608965 A	9/2020
CN	111664087 A	9/2020
CN	111677476 A	9/2020
CN	111677647 A	9/2020
CN	111692064 A	9/2020
CN	111692065 A	9/2020
CN	211384571	9/2020
CN	211397553	9/2020
CN	211397677	9/2020
CN	211500955	9/2020
CN	211524765	9/2020
DE	4004854	8/1991
DE	4241614	6/1994
DE	102009022859	12/2010
DE	102012018825	3/2014
DE	102013111655	12/2014
DE	102015103872	10/2015
DE	102013114335	12/2020
EP	0835983	4/1998
EP	1378683	1/2004
EP	2143916	1/2010
EP	2613023	7/2013
EP	3095989	11/2016
EP	3211766	8/2017
EP	3049642	4/2018
EP	3354866	8/2018
EP	3075946	5/2019
FR	2795774	6/1999
GB	474072	10/1937
GB	1438172	6/1976
JP	857135212	2/1984
KR	20020026398	4/2002
RU	13562	4/2000
WO	1993020328	10/1993
WO	2006025886	3/2006
WO	2009023042	2/2009
WO	20110133821	10/2011
WO	2012139380	10/2012
WO	2013158822	10/2013
WO PCT/CN	2012/074945	11/2013
WO	2013185399	12/2013
WO	2015158020	10/2015
WO	2016014476	1/2016
WO	2016033983	3/2016
WO	2016078181	5/2016
WO	2016101374	6/2016
WO	2016112590	7/2016
WO	2016/186790	11/2016
WO	2017123656 A	7/2017
WO	2017146279	8/2017
WO	2017213848	12/2017
WO	2018031029	2/2018
WO	2018038710	3/2018
WO	2018044293	3/2018
WO	2018044307	3/2018
WO	2018071738	4/2018
WO	2018101909	6/2018
WO	2018101912	6/2018
WO	2018106210	6/2018
WO	2018106225	6/2018
WO	2018106252	6/2018

WO	2018/132106	7/2018
WO	2018156131	8/2018
WO	2018075034	10/2018
WO	2018187346	10/2018
WO	2018031031	2/2019
WO	2019045691	3/2019
WO	2019046680	3/2019
WO	2019060922	3/2019
WO	2019117862	6/2019
WO	2019126742	6/2019
WO	2019147601	8/2019
WO	2019169366	9/2019
WO	2019195651	10/2019
WO	2019200510	10/2019
WO	2019210417	11/2019
WO	2020018068	1/2020
WO	2020046866	3/2020
WO	2020072076	4/2020
WO	2020076569	4/2020
WO	2020097060	5/2020
WO	2020104088	5/2020
WO	2020131085	6/2020
WO	2020211083	10/2020
WO	2020211086	10/2020
WO	2021038604	3/2021
WO	2021041783	3/2021

OTHER PUBLICATIONS

Final written decision of PGR2021-00102 dated Feb. 6, 2023.

Final written decision of PGR2021-00103 dated Feb. 6, 2023.

AFGlobal Corporation, Durastim Hydraulic Fracturing Pump, A Revolutionary Design for Continuous Duty Hydraulic Fracturing, 2018.

SPM® QEM 5000 E-Frac Pump Specification Sheet, Weir Group (2019) ("Weir 5000").

Green Field Energy Services Natural Gas Driven Turbine Frac Pumps HHP Summit Presentation, Yumpu (Sep. 2012), <https://www.yumpu.com/en/document/read/49685291/turbine-frac-pump-assembly-hhp> ("Green Field").

Dowell B908 "Turbo-Jet" Operator's Manual.

Jereh Debut's Super-power Turbine Fracturing Pump, Leading the Industrial Revolution, Jereh Oilfield Services Group (Mar. 19, 2014), <https://www.prnewswire.com/news-releases/jereh-debuts-super-power-turbine-fracturing-pump-leading-the-industrial-revolution-250992111.html>.

Jereh Apollo 4500 Turbine Frac Pumper Finishes Successful Field Operation in China, Jereh Group (Feb. 13, 2015), as available on Apr. 20, 2015, <https://web.archive.org/web/20150420220625/https://www.prnewswire.com/news-releases/jereh-apollo-4500-turbine-frac-pumper-finishes-successful-field-operation-in-china-300035829.html>. 35% Economy Increase, Dual-fuel System Highlighting Jereh Apollo Frac Pumper, Jereh Group (Apr. 13, 2015), <https://www.jereh.com/en/news/press-release/news-detail-7345.htm>.

Hydraulic Fracturing: Gas turbine proves successful in shale gas field operations, Vericor (2017), <https://www.vericor.com/wp-content/uploads/2020/02/7.-Fracing-4500hp-Pump-China-En.pdf> ("Vericor Case Study").

Jereh Apollo Turbine Fracturing Pumper Featured on China Central Television, Jereh Group (Mar. 9, 2018), <https://www.jereh.com/en/news/press-release/news-detail-7267.htm>.

Jereh Unveiled New Electric Fracturing Solution at OTC 2019, Jereh Group (May 7, 2019), as available on May 28, 2019, <https://web.archive.org/web/20190528183906/https://www.prnewswire.com/news-releases/jereh-unveiled-new-electric-fracturing-solution-at-otc-2019-300845028.html>.

Jereh Group, Jereh Fracturing Unit, Fracturing Spread, YouTube (Mar. 30, 2015), <https://www.youtube.com/watch?v=PlkDbU5dE0o>. Transcript of Jereh Group, Jereh Fracturing Unit, Fracturing Spread, YouTube (Mar. 30, 2015).

Jereh Group, Jereh Fracturing Equipment. YouTube (Jun. 8, 2015), <https://www.youtube.com/watch?v=m0vMiq84P4Q>.

Transcript of Jereh Group, Jereh Fracturing Equipment, YouTube (Jun. 8, 2015), <https://www.youtube.com/watch?v=m0vMiq84P4Q>.

(56)

References Cited**OTHER PUBLICATIONS**

- Ferdinand P. Beer et al., *Mechanics of Materials* (6th ed. 2012).
- Weir Oil & Gas Introduces Industry's First Continuous Duty 5000-Horsepower Pump, Weir Group (Jul. 25, 2019), <https://www.global.weir/newsroom/news-articles/weir-oil-and-gas-introduces-industrys-first-continuous-duty-5000-horsepower-pump/>.
- 2012 High Horsepower Summit Agenda, Natural Gas for High Horsepower Applications (Sep. 5, 2012).
- Review of HHP Summit 2012, Gladstein, Neandross & Associates <https://www.gladstein.org/gna-conferences/high-horsepower-summit-2012/>.
- Green Field Energy Services Deploys Third New Hydraulic Fracturing System, Green Field Energy Services, Inc. (Jul. 11, 2012), <https://www.prnewswire.com/news-releases/green-field-energy-services-deploys-third-new-hydraulic-fracturing-spread-162113425>.
- Karen Boman, Turbine Technology Powers Green Field Multi-Fuel Frack Pump, Rigzone (Mar. 7, 2015), as available on Mar. 14, 2015, https://web.archive.org/web/20150314203227/https://www.rigzone.com/news/oil-gas/a/124883/Turbine_Technology_Powers_Green_Field_MultiFuel_Frack_Pump.
- "Turbine Frac Units," WMD Squared (2012), <https://wmdsquared.com/work/gfes-turbine-frac-units/>.
- Leslie Turj, Green Field asset sale called 'largest disposition industry has seen,' The Insider Media (Mar. 19, 2014), <http://theind.com/article-16497-green-field-asset-sale-called-%E2%80%98largest-disposition-industry-has-seen%60.html>.
- "Honghua developing new-generation shale-drilling rig, plans testing of frac pump"; Katherine Scott; Drilling Contractor May 23, 2013; accessed at <https://www.drillingcontractor.org/honghua-developing-new-generation-shale-drilling-rig-plans-testing-of-frac-pump-23278>.
- Europump and Hydraulic Institute, *Variable Speed Pumping: A Guide to Successful Applications*, Elsevier Ltd, 2004.
- Capstone Turbine Corporation, Capstone Receives Three Megawatt Order from Large Independent Oil & Gas Company in Eagle Ford Shale Play, Dec. 7, 2010.
- Wikipedia, Westinghouse Combustion Turbine Systems Division, https://en.wikipedia.org/wiki/Westinghouse_Combustion_Turbine_Systems_Division, circa 1960.
- Wikipedia, Union Pacific GTELs, https://en.wikipedia.org/wiki/Union_Pacific_GTEs, circa 1950.
- HCI JET Frac, Screenshots from YouTube, Dec. 11, 2010. <https://www.youtube.com/watch?v=6HjXkdbFaFQ>.
- Afd Petroleum Ltd., Automated Hot Zone, Frac Refueling System, Dec. 2018.
- Eygun, Christiane, et al., URTEC: 2687987, Mitigating Shale Gas Developments Carbon Footprint: Evaluating and Implementing Solutions in Argentina, Copyright 2017, Unconventional Resources Technology Conference.
- Walzel, Brian, Hart Energy, Oil, Gas Industry Discovers Innovative Solutions to Environmental Concerns, Dec. 10, 2018.
- Frac Shack, Bi-Fuel FracFueller brochure, 2011.
- Pettigrew, Dana, et al., High Pressure Multi-Stage Centrifugal Pump for 10,000 psi Frac Pump—HPHPS FRAC Pump, Copyright 2013, Society of Petroleum Engineers, SPE 166191.
- Elle Seybold, et al., Evolution of Dual Fuel Pressure Pumping for Fracturing: Methods, Economics, Field Trial Results and Improvements in Availability of Fuel, Copyright 2013, Society of Petroleum Engineers, SPE 166443.
- Wallace, E.M., Associated Shale Gas: From Flares to Rig Power, Copyright 2015, Society of Petroleum Engineers, SPE-173491-MS.
- Williams, C.W. (Gulf Oil Corp. Odessa Texas), The Use of Gas-turbine Engines in an Automated High-Pressure Water-injection Stations; American Petroleum Institute; API-63-144 (Jan. 1, 1963).
- Neal, J.C. (Gulf Oil Corp. Odessa Texas), Gas Turbine Driven Centrifugal Pumps for High Pressure Water Injection; American Institute of Mining, Metallurgical and Petroleum Engineers, Inc.; SPE-1888 (1967).
- Porter, John A. (SOLAR Division International Harvester Co.), Modern Industrial Gas Turbines for the Oil Field; American Petroleum Institute; Drilling and Production Practice; API-67-243 (Jan. 1, 1967).
- Cooper et al., Jet Frac Porta-Skid—A New Concept in Oil Field Service Pump Equipments[sic]; Halliburton Services; SPE-2706 (1969).
- Ibragimov, É.S., Use of gas-turbine engines in oil field pumping units; Chem Petrol Eng; (1994) 30: 530. <https://doi.org/10.1007/BF01154919>. (Translated from Khimicheskaya i Neftyanoe Mashinostroenie, No. 11, pp. 24-26, Nov. 1994.).
- Kas'yanov et al., Application of gas-turbine engines in pumping units complexes of hydraulic fracturing of oil and gas reservoirs; Exposition Oil & Gas; (Oct. 2012) (published in Russian).
- American Petroleum Institute. API 674: Positive Displacement Pumps—Reciprocating. 3rd ed. Washington, DC: API Publishing Services, 2010.
- American Petroleum Institute. API 616: Gas Turbines for the Petroleum, Chemical, and Gas Industry Services. 5th ed. Washington, DC: API Publishing Services, 2011.
- Karassik, Igor, Joseph Messina, Paul Cooper, and Charles Heald. Pump Handbook. 4th ed. New York: McGraw-Hill Education, 2008.
- Weir SPM. Weir SPM General Catalog: Well Service Pumps, Flow Control Products, Manifold Trailers, Safety Products, Post Sale Services. Ft. Worth, TX: Weir Oil & Gas. May 28, 2016. <https://www.pumpfundamentals.com/pumpdatabase2/weir-spm-general.pdf>.
- The Weir Group, Inc. Weir SPM Pump Product Catalog. Ft. Worth, TX: S.P.M. Flow Control, Inc. Oct. 30, 2017. https://manage.global.weir/assets/files/product%20brochures/SPM_2P140706_Pump_Product_Catalogue_View.pdf.
- Shandong Saigao Group Corporation. Q4 (5W115) Quintuplex Plunger Pump. Jinan City, Shandong Province, China: Saigao. Oct. 20, 2014. <https://www.saigaogroup.com/product/q400-5w115-quintuplex-plunger-pump.html>.
- Marine Turbine. Turbine Powered Frac Units. Franklin, Louisiana: Marine Turbine Technologies, 2020.
- Rotating Right. Quintuplex Power Pump Model Q700. Edmonton, Alberta, Canada: Weatherford International Ltd. <https://www.rotatingright.com/pdf/weatherford/RR%2026-Weatherford%20Model%20Q700.pdf>, 2021.
- CanDyne Pump Services, Inc. Weatherford Q700 Pump. Calgary, Alberta, Canada: CanDyne Pump Services. Aug. 15, 2015. <http://candyne.com/wp-content/uploads/2014/10/181905-94921.q700-quintuplex-pump.pdf>.
- Arop, Julius Bankong. Geomechanical review of hydraulic fracturing technology. Thesis (M. Eng.). Cambridge, MA: Massachusetts Institute of Technology, Dept. of Civil and Environmental Engineering. Oct. 29, 2013. <https://dspace.mit.edu/handle/1721.1/82176>.
- ResearchGate, Answer by Byron Woolridge, found at https://www.researchgate.net/post/How_can_we_improve_the_efficiency_of_the_gas_turbine_cycles, Jan. 1, 2013.
- Filipović, Ivan, Preliminary Selection of Basic Parameters of Different Torsional Vibration Dampers Intended for use in Medium-Speed Diesel Engines, Transactions of Famen XXXVI-3 (2012).
- Marine Turbine Technologies, 1 MW Power Generation Package, <http://marineturbine.com/power-generation>, 2017.
- Business Week: Fiber-optic cables help fracking, cablinginstall.com. Jul. 12, 2013. <https://www.cablinginstall.com/cable/article/16474208/businessweek-fiber-optic-cables-help-fracking>.
- Fracking companies switch to electric motors to power pumps, iadd-intl.org. Jun. 27, 2019. <https://www.iadd-intl.org/articles/fracking-companies-switch-to-electric-motors-to-power-pumps/>.
- The Leader in Frac Fueling, suncoastresources.com. Jun. 29, 2015. <https://web.archive.org/web/20150629220609/https://www.suncoastresources.com/oilfield/fueling-services/>.
- Mobile Fuel Delivery, atlasoil.com. Mar. 6, 2019. <https://www.atlasoil.com/nationwide-fueling/onsite-and-mobile-fueling>.
- Frac Tank Hose (FRAC), 4starhose.com. Accessed: Nov. 10, 2019. http://www.4starhose.com/product/frac_tank_hose_frac.aspx.

(56)

References Cited**OTHER PUBLICATIONS**

- Plos One, Dynamic Behavior of Reciprocating Plunger Pump Discharge Valve Based on Fluid Structure Interaction and Experimental Analysis. Oct. 21, 2015.
- FMC Technologies, Operation and Maintenance Manual, L06 Through L16 Triplex Pumps Doc No. OMM50000903 Rev: E p. 1 of 66. Aug. 27, 2009.
- Gardner Denver Hydraulic Fracturing Pumps GD 3000 <https://www.gardnerdenver.com/en-us/pumps/triplex-fracking-pump-gd-3000>.
- Lekontsev, Yu M., et al. "Two-side sealer operation." *Journal of Mining Science* 49.5 (2013): 757-762.
- Tom Hausfeld, GE Power & Water, and Eldon Schelske, Evolution Well Services, TM2500+ Power for Hydraulic Fracturing.
- FTS International's Dual Fuel Hydraulic Fracturing Equipment Increases Operational Efficiencies, Provides Cost Benefits, Jan. 3, 2018.
- CNG Delivery, Fracturing with natural gas, dual-fuel drilling with CNG, Aug. 22, 2019.
- PbNG, Natural Gas Fuel for Drilling and Hydraulic Fracturing, Diesel Displacement / Dual Fuel & Bi-Fuel, May 2014.
- Integrated Flow, Skid-mounted Modular Process Systems, Jul. 15, 2017, <https://ifsolutions.com/why-modular/>.
- Cameron, A Schlumberger Company, Frac Manifold Systems, 2016.
- ZSI-Foster, Energy | Solar | Fracking | Oil and Gas, Aug. 2020, <https://www.zsi-foster.com/energy-solar-fracking-oil-and-gas.html>.
- JBG Enterprises, Inc., WS-Series Blowout Prevention Safety Coupling—Quick Release Couplings, Sep. 11, 2015, <http://www.jgbhose.com/products/WS-Series-Blowout-Prevention-Safety-Coupling.asp>.
- Halliburton, Vessel-based Modular Solution (VMS), 2015.
- Chun, M. K., H. K. Song, and R. Lallemand. "Heavy duty gas turbines in petrochemical plants: Samsung's Daesan plant (Korea) beats fuel flexibility records with over 95% hydrogen in process gas." *Proceedings of PowerGen Asia Conference*, Singapore. 1999.
- Wolf, Jürgen J., and Marko A. Perkavec. "Safety Aspects and Environmental Considerations for a 10 MW Cogeneration Heavy Duty Gas Turbine Burning Coke Oven Gas with 60% Hydrogen Content." *ASME 1992 International Gas Turbine and Aeroengine Congress and Exposition*. American Society of Mechanical Engineers Digital Collection, 1992.
- Ginter, Timothy, and Thomas Bouvay. "Uprate options for the MS7001 heavy duty gas turbine." *GE paper GER-3808C*, GE Energy 12 (2006).
- Chaichan, Miqdam Tariq. "The impact of equivalence ratio on performance and emissions of a hydrogen-diesel dual fuel engine with cooled exhaust gas recirculation." *International Journal of Scientific & Engineering Research* 6.6 (2015): 938-941.
- Ecob, David J., et al. "Design and Development of a Landfill Gas Combustion System for the Typhoon Gas Turbine." *ASME 1996 International Gas Turbine and Aeroengine Congress and Exhibition*. American Society of Mechanical Engineers Digital Collection, 1996.
- II-VI Marlow Industries, Thermoelectric Technologies in Oil, Gas, and Mining Industries, blog.marlow.com (Jul. 24, 2019).
- B.M. Mahlalela, et al., "Electric Power Generation Potential Based on Waste Heat and Geothermal Resources in South Africa," *pangea.stanford.edu* (Feb. 11, 2019).
- Department of Energy, United States of America, *The Water-Energy Nexus: Challenges and Opportunities* ourenergypolicy.org (Jun. 2014).
- Ankit Tiwari, Design of a Cooling System for a Hydraulic Fracturing Equipment, The Pennsylvania State University, The Graduate School, College of Engineering, 2015.
- Jp Yadav et al., Power Enhancement of Gas Turbine Plant by Intake Air Fog Cooling, Jun. 2015.
- Mee Industries: Inlet Air Fogging Systems for Oil, Gas and Petrochemical Processing, Verdict Media Limited Copyright 2020.
- M. Ahmadzadehtalatpeh et al. Performance enhancement of gas turbine units by retrofitting with inlet air cooling technologies (IACTs): an hour-by-hour simulation study, *Journal of the Brazilian Society of Mechanical Sciences and Engineering*, Mar. 2020.
- Advances in Popular Torque-Link Solution Offer OEMs Greater Benefit, Jun. 21, 2018.
- Emmanuel Akita et al., Mewbourne College of Earth & Energy, Society of Petroleum Engineers; Drilling Systems Automation Technical Section (DSATS); 2019.
- PowerShelter Kit II, nooutage.com, Sep. 6, 2019.
- EMPengineering.com, HEMP Resistant Electrical Generators / Hardened Structures HEMP/GMD Shielded Generators, Virginia, Nov. 3, 2012.
- Blago Minovski, Coupled Simulations of Cooling and Engine Systems for Unsteady Analysis of the Benefits of Thermal Engine Encapsulation, Department of Applied Mechanics, Chalmers University of Technology Goteborg, Sweden 2015.
- J. Porteiro et al., Feasibility of a new domestic CHP trigeneration with heat pump: II. Availability analysis. Design and development, *Applied Thermal Engineering* 24 (2004) 1421-1429.
- ISM, What is Cracking Pressure, 2019.
- Swagelok, The right valve for controlling flow direction? Check, 2016.
- Technology.org, Check valves how do they work and what are the main type, 2018.
- De Gevigney et al., "Analysis of no-load dependent power losses in a planetary gear train by using thermal network method", *International Gear Conference 2014*: Aug. 26-28, 2014, Lyon, pp. 615-624.
- Special-Purpose Couplings for Petroleum, Chemical, and Gas Industry Services, API Standard 671 (4th Edition) (2010).
- The Application of Flexible Couplings for Turbomachinery, Jon R. Mancuso et al., *Proceedings of the Eighteenth Turbomachinery Symposium* (1989).
- Pump Control With Variable Frequency Drives, Kevin Tory, *Pumps & Systems: Advances in Motors and Drives*, Reprint from Jun. 2008.
- Fracture Design and Stimulation, Mike Eberhard, P.E., *Well Construction & Operations Technical Workshop* In support of the Epa Hydraulic Fracturing Study, Mar. 10-11, 2011.
- General Purpose vs. Special Purpose Couplings, Jon Mancuso, *Proceedings of the Twenty-Third Turbomachinery Symposium* (1994).
- Overview of Industry Guidance/Best Practices on Hydraulic Fracturing (HF), American Petroleum Institute, © 2012.
- API Member Companies, American Petroleum Institute, WaybackMachine Capture, <https://web.archive.org/web/20130424080625/http://api.org/globalitems/globalheaderpages/membership/api-member-companies>, accessed Jan. 4, 2021.
- API's Global Industry Services, American Petroleum Institute, © Aug. 2020.
- About API, American Petroleum Institute, <https://www.api.org/about>, accessed Dec. 30, 2021.
- About API, American Petroleum Institute, WaybackMachine Capture, <https://web.archive.org/web/20110422104346/http://api.org/aboutapi/>, captured Apr. 22, 2011.
- Publications, American Petroleum Institute, WaybackMachine Capture, <https://web.archive.org/web/20110427043936/http://www.api.org/80/Publications/>, captured Apr. 27, 2011.
- Procedures for Standards Development, American Petroleum Institute, Third Edition (2006).
- WorldCat Library Collections Database Records for API Standard 671 and API Standard 674, https://www.worldcat.org/title/positive-displacement-pumps-reciprocating/oclc/858692269&referer=brief_results, accessed Dec. 30, 2021; and https://www.worldcat.org/title/special-purpose-couplings-for-petroleum-chemical-and-gas-industry-services/oclc/871254217&referer=brief_results, accessed Dec. 22, 2021.
- 2011 Publications and Services, American Petroleum Institute (2011). Standards, American Petroleum Institute, WaybackMachine Capture, <https://web.archive.org/web/20110207195046/http://www.api.org/Standards/>, captured Feb. 7, 2011; and <https://web.archive.org/web/20110204112554/http://global.ihs.com/?RID=API1>, captured Feb. 4, 2011.
- IHS Markit Standards Store, https://global.ihs.com/doc_detail.cfm?document_name=API%20STD%20674&item_s_key=00010672#doc-detail-history-anchor, accessed Dec. 30, 2021; and https://global.ihs.com/doc_detail.cfm?&input_doc_number=671&input_doc_title=

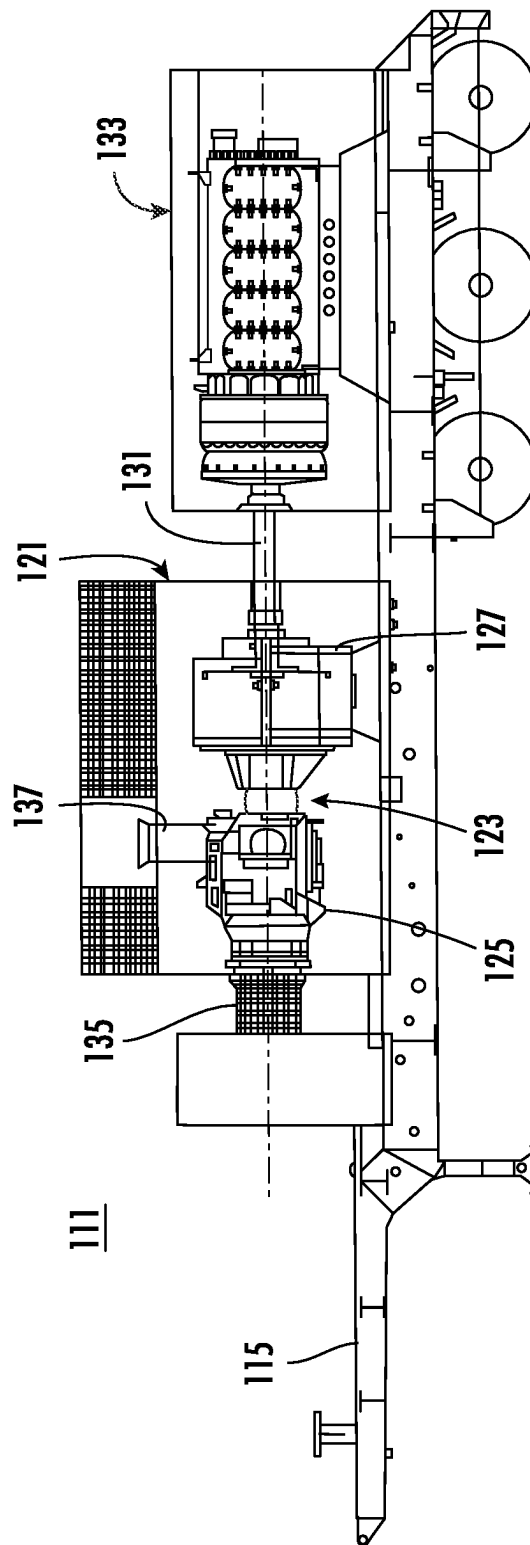
(56)

References Cited

OTHER PUBLICATIONS

&document_name=API%20STD%20671&item_s_key=00010669
&item_key_date=890331&origin=DSSC, accessed Dec. 30, 2021.
Dziubak, Tadeusz, "Experimental Studies of Dust Suction Irregularity from Multi-Cyclone Dust Collector of Two-Stage Air Filter",
Energies 2021, 14, 3577, 28 pages.

* cited by examiner







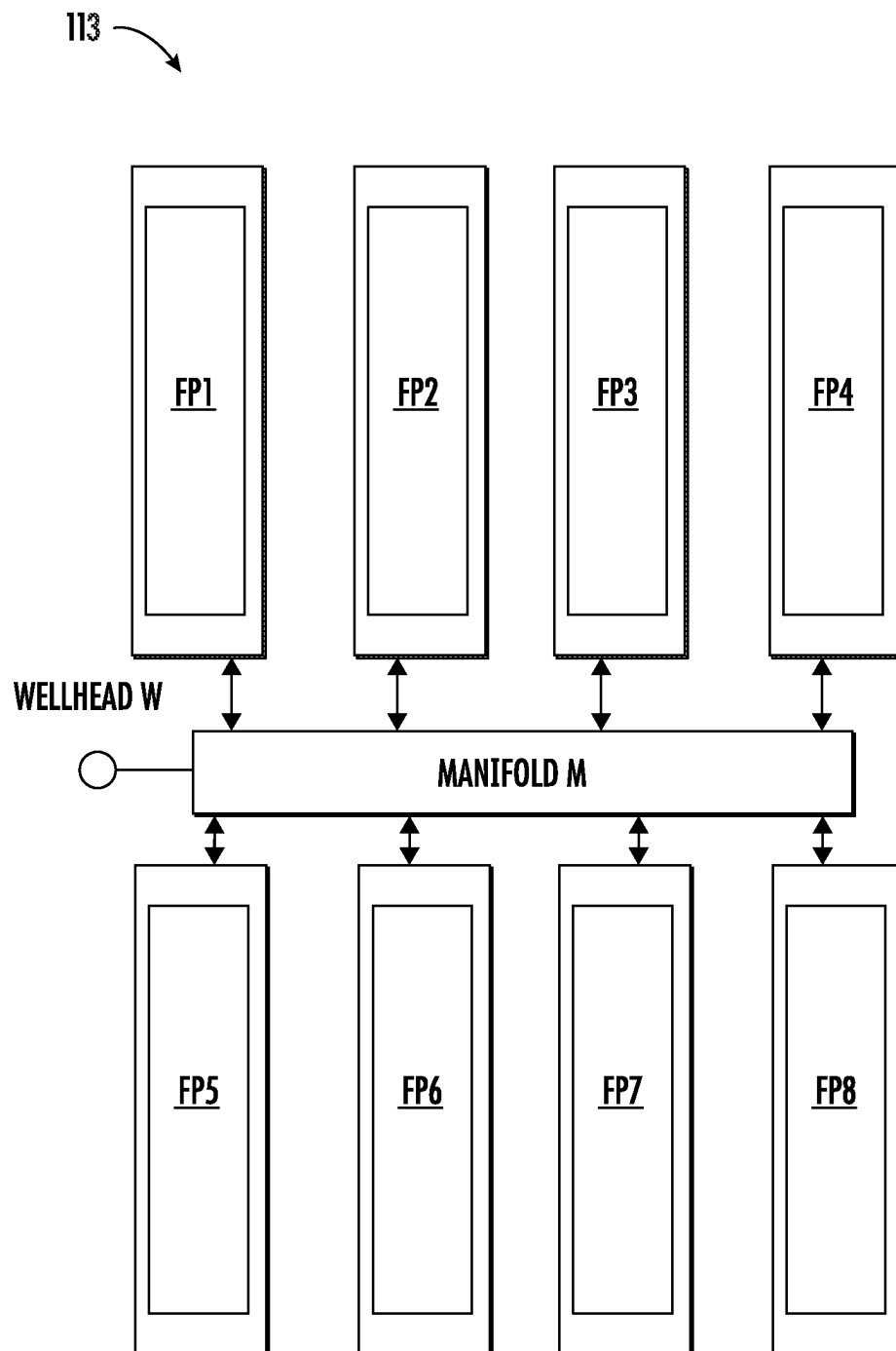
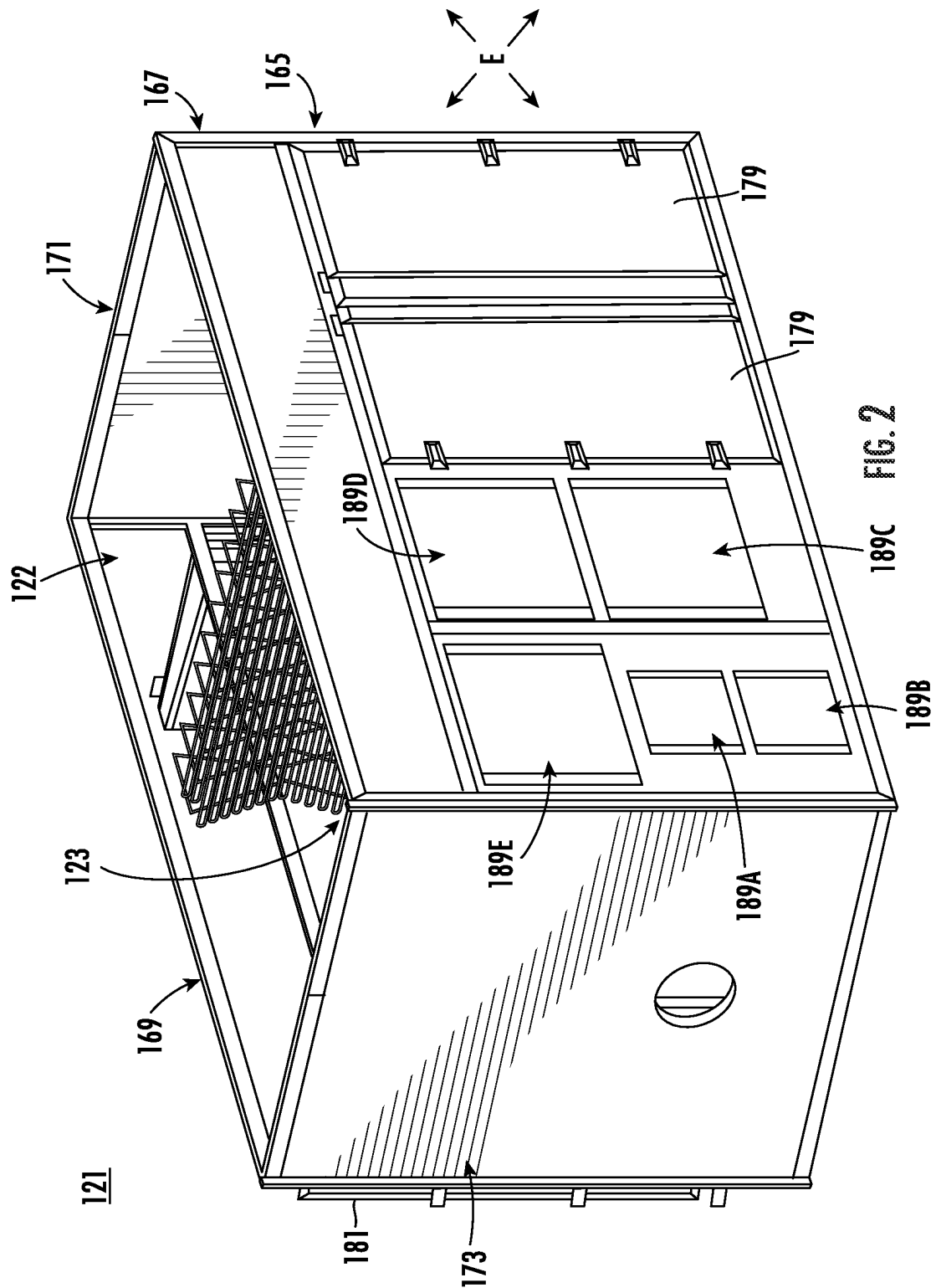


FIG. 1B



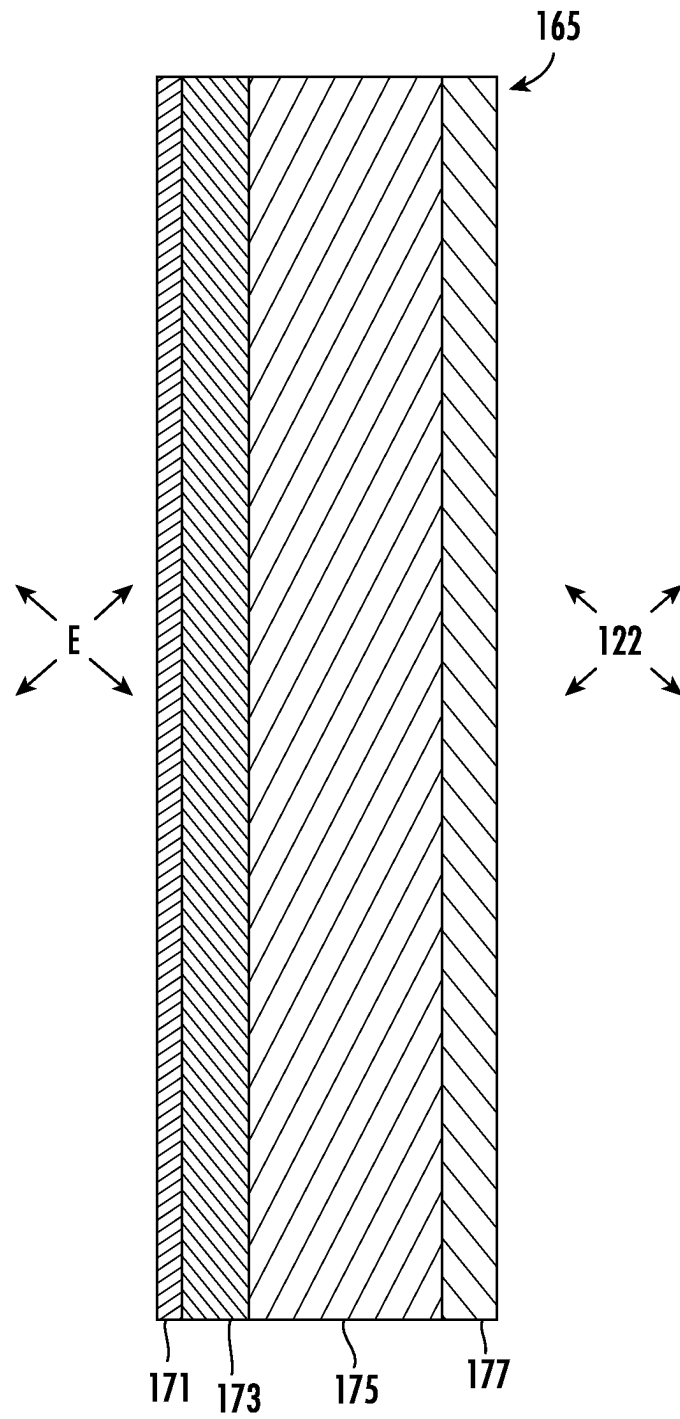


FIG. 3

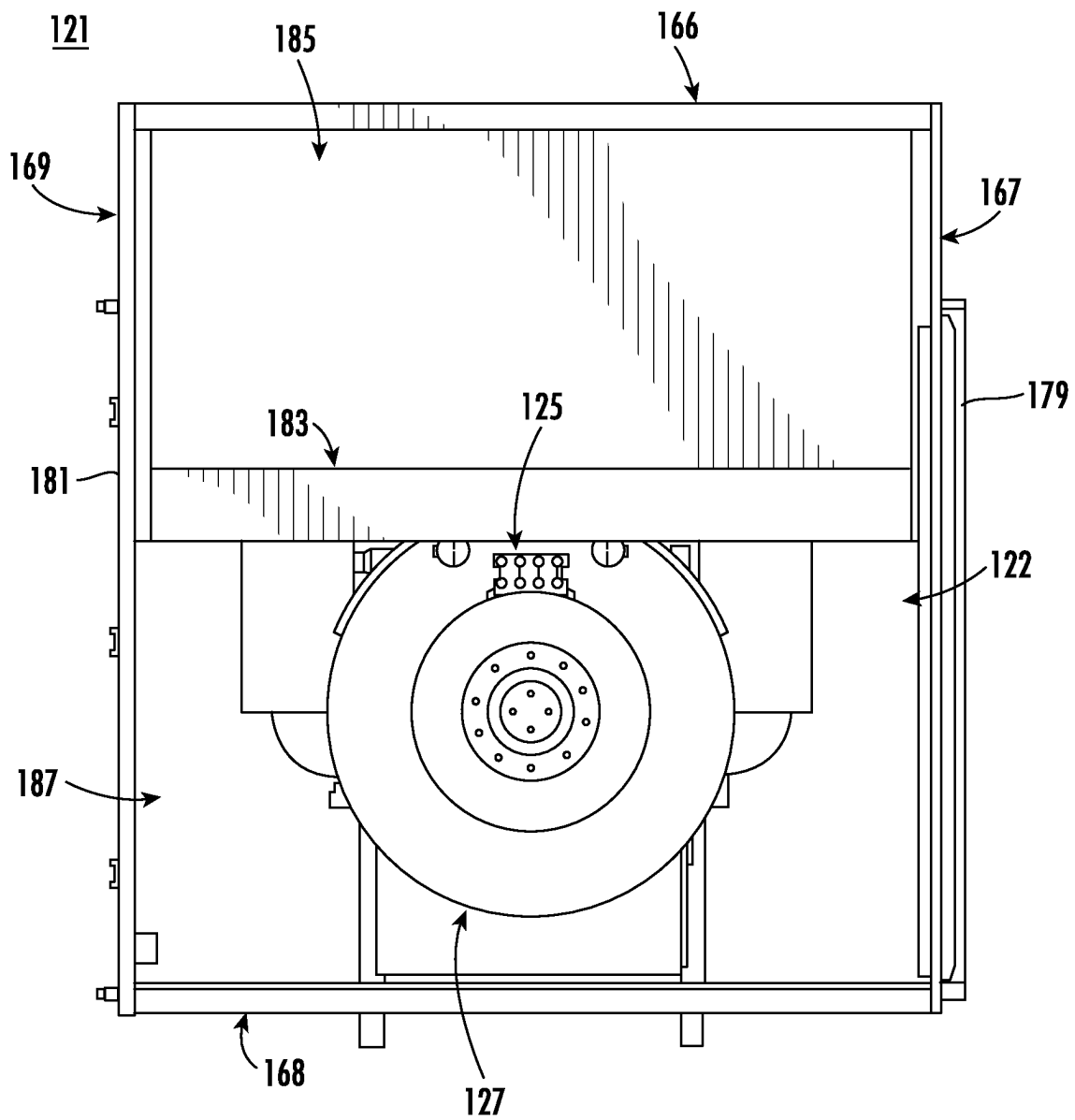


FIG. 4

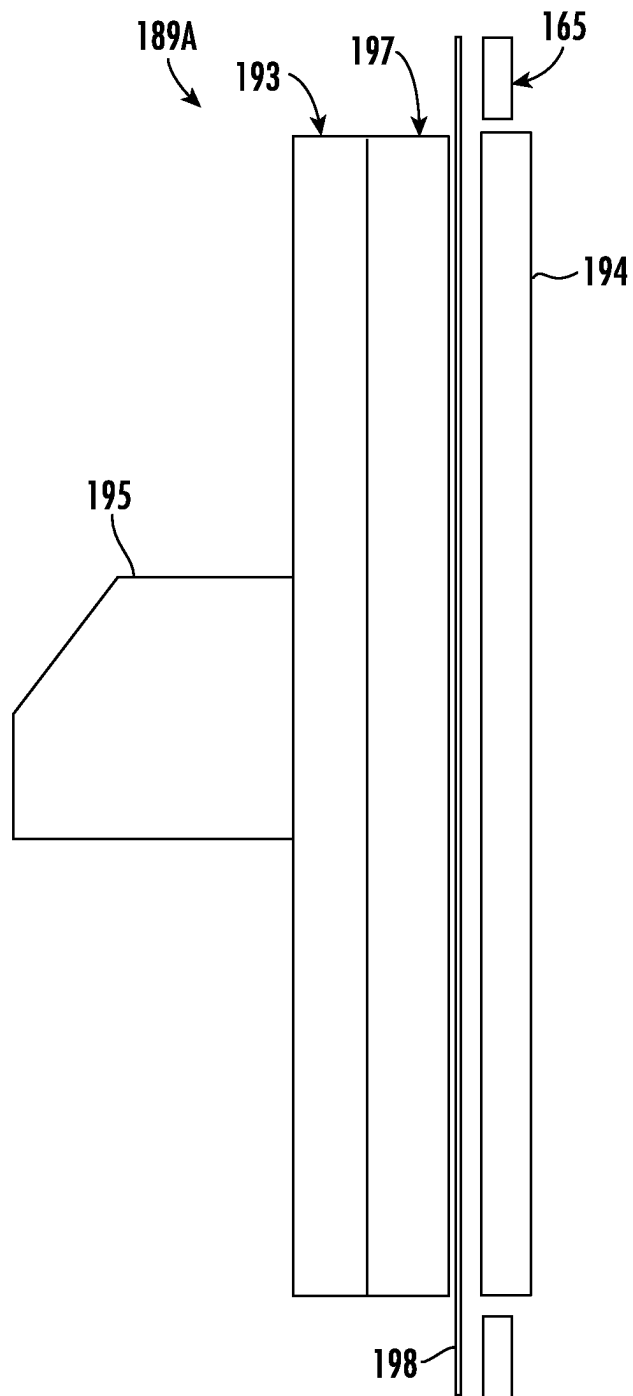


FIG. 5

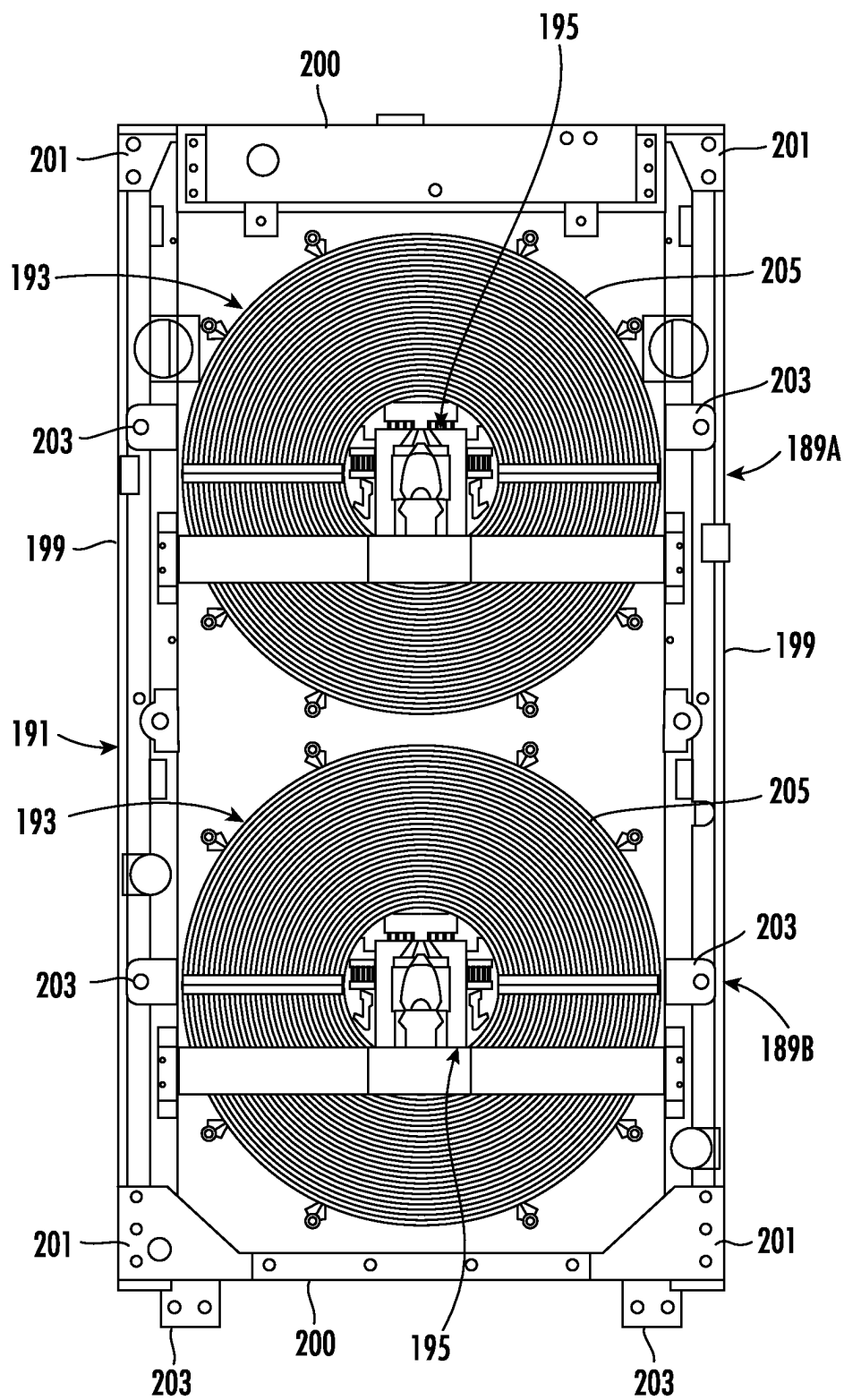


FIG. 6

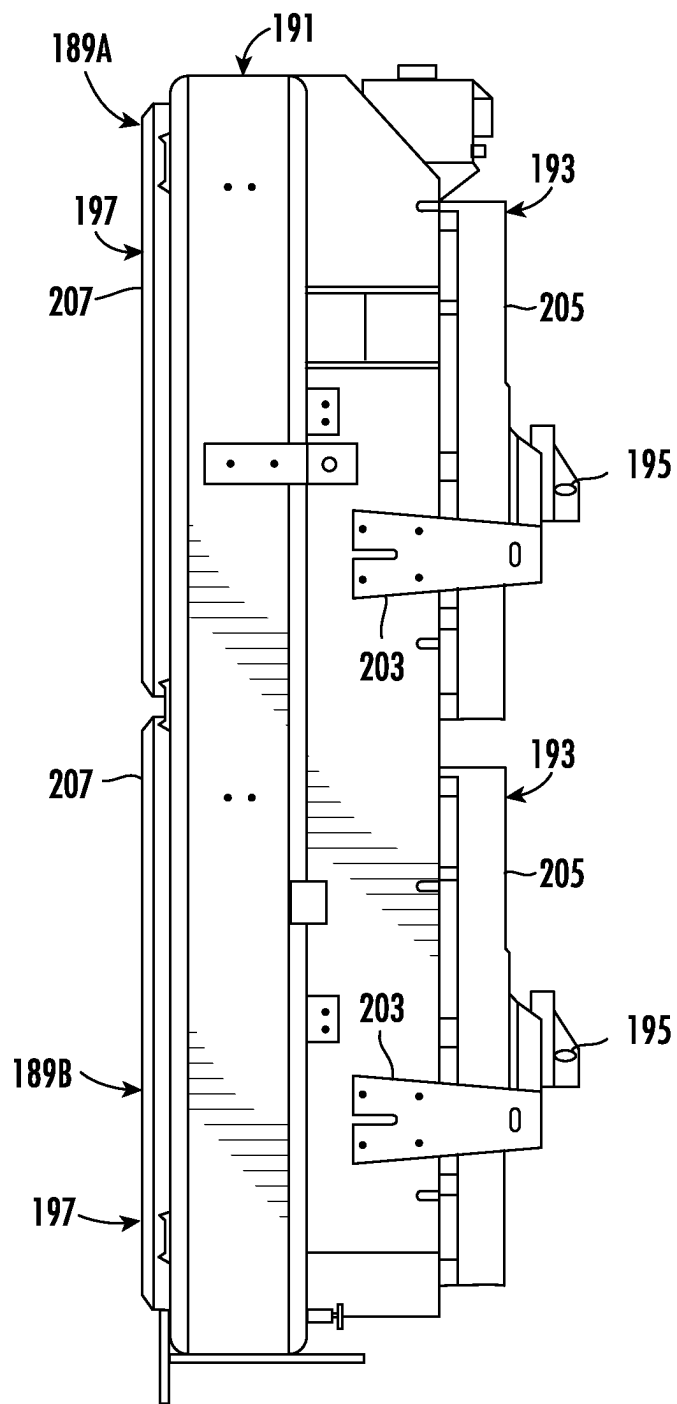


FIG. 7

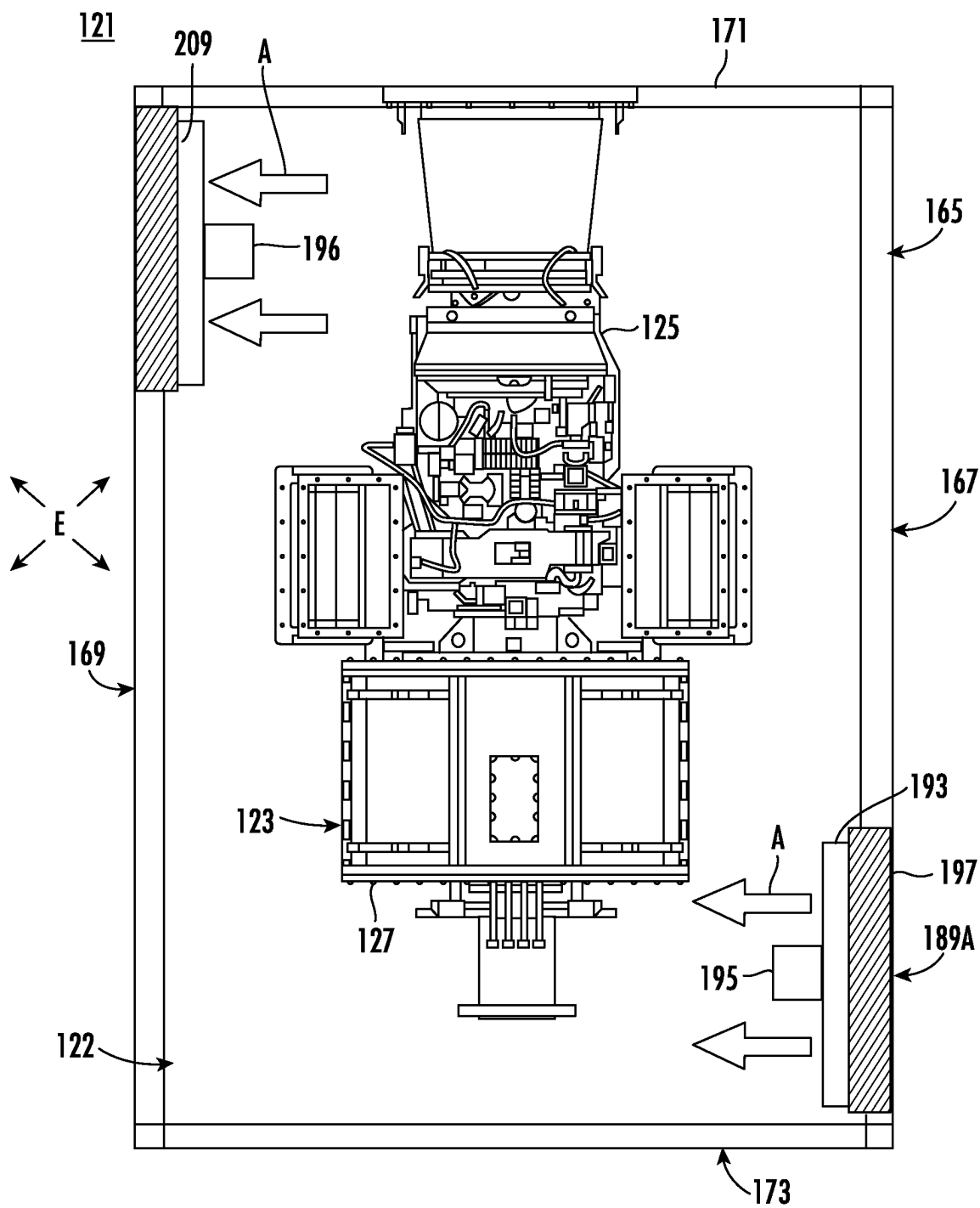
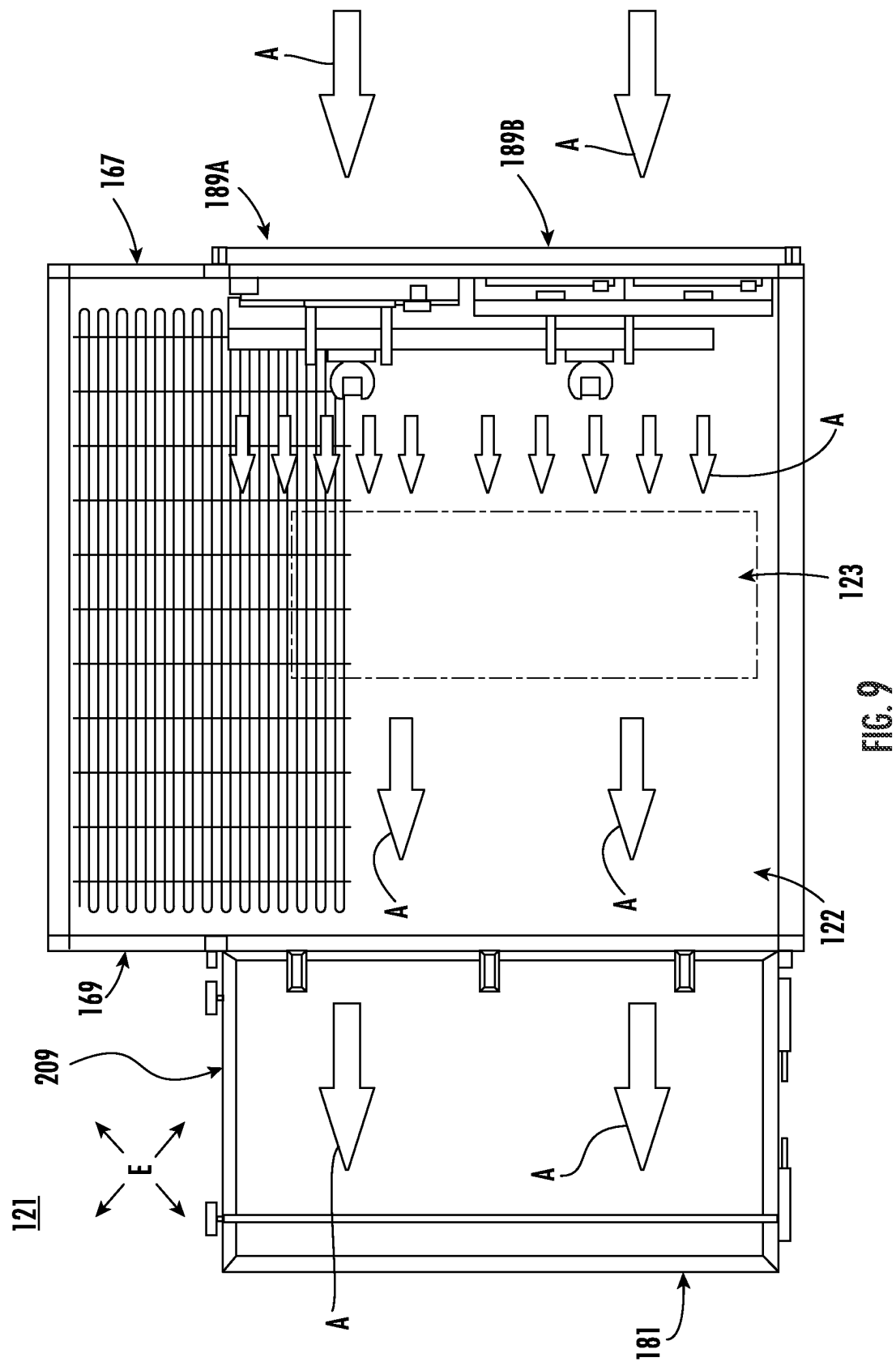
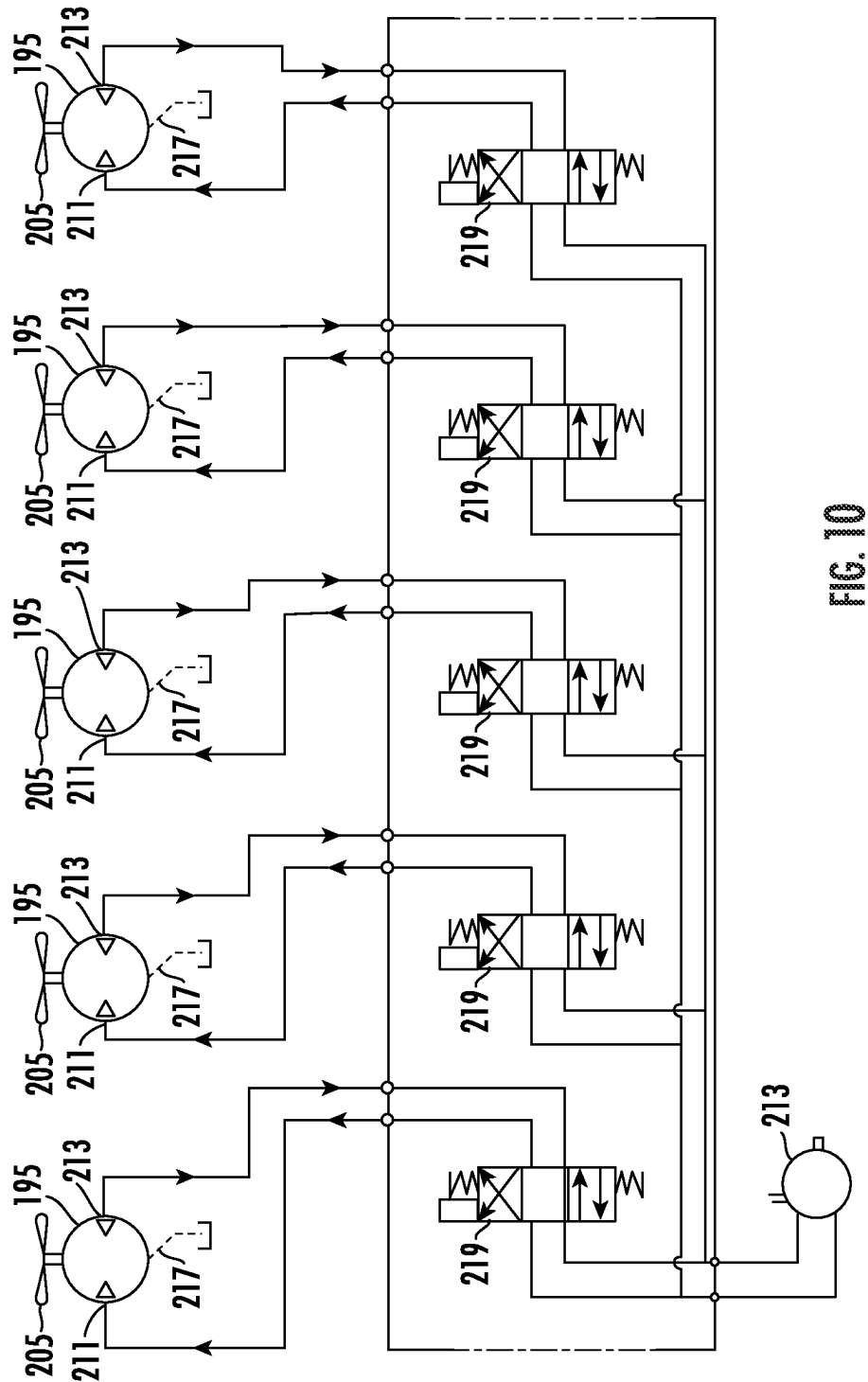


FIG. 8





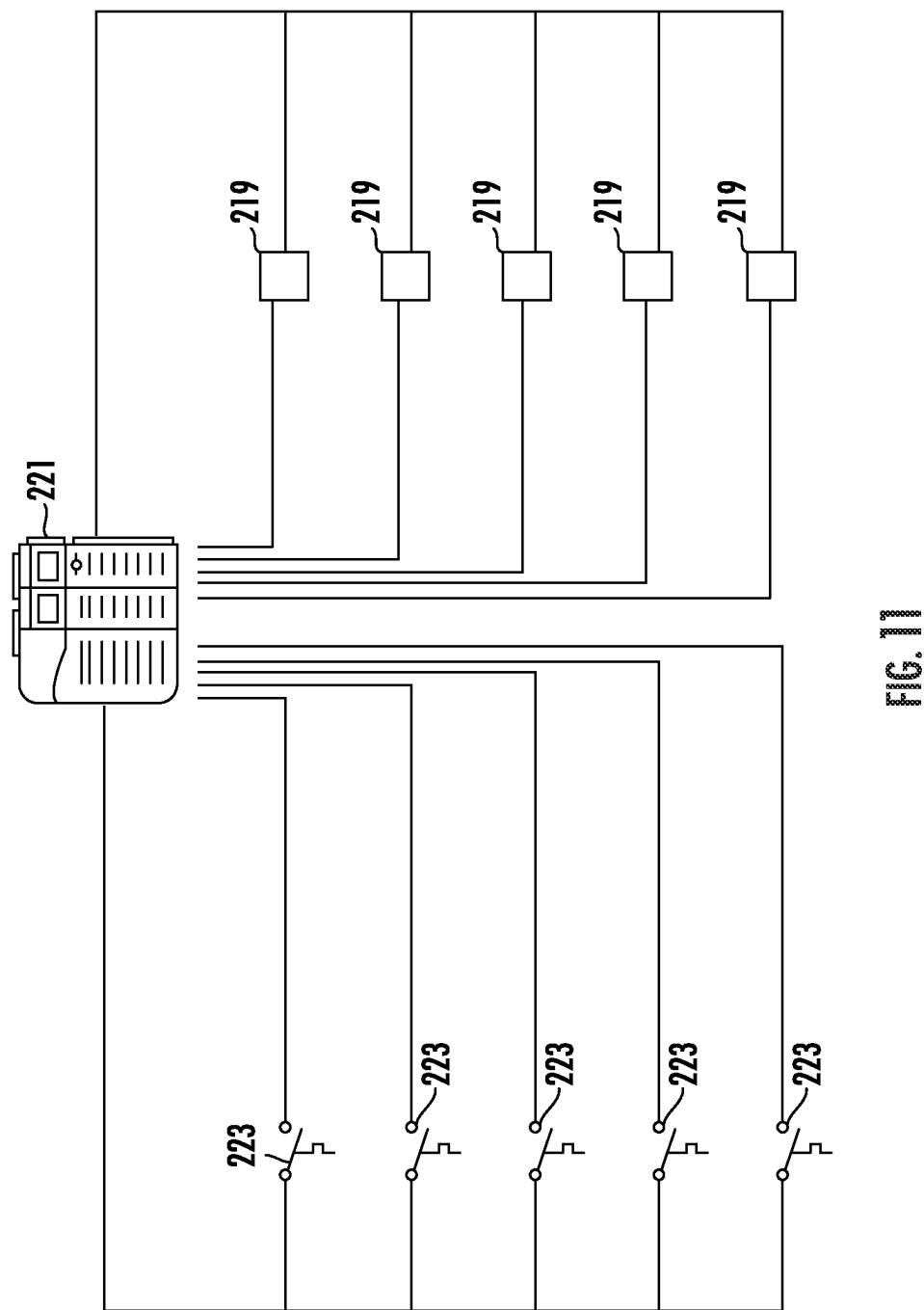


FIG. 1

1

ENCLOSURE ASSEMBLY FOR ENHANCED COOLING OF DIRECT DRIVE UNIT AND RELATED METHODS

PRIORITY CLAIM

This is a continuation of U.S. Non-Provisional application Ser. No. 18/096,927, filed Jan. 13, 2023, titled "ENCLOSURE ASSEMBLY FOR ENHANCED COOLING OF DIRECT DRIVE UNIT AND RELATED METHODS," which is a continuation of U.S. Non-Provisional application Ser. No. 17/444,485, filed Aug. 5, 2021, titled "ENCLOSURE ASSEMBLY FOR ENHANCED COOLING OF DIRECT DRIVE UNIT AND RELATED METHODS," now U.S. Pat. No. 11,627,683, issued Apr. 11, 2023, which is a continuation of U.S. Non-Provisional application Ser. No. 17/356,063, filed Jun. 23, 2021, titled "ENCLOSURE ASSEMBLY FOR ENHANCED COOLING OF DIRECT DRIVE UNIT AND RELATED METHODS," now U.S. Pat. No. 11,129,295, issued Sep. 21, 2021, which is a divisional of U.S. Non-Provisional application Ser. No. 17/302,039, filed Apr. 22, 2021, titled "ENCLOSURE ASSEMBLY FOR ENHANCED COOLING OF DIRECT DRIVE UNIT AND RELATED METHODS," now U.S. Pat. No. 11,109,508, issued Aug. 31, 2021, which claims priority to and the benefit of, under 35 U.S.C. § 119(e), U.S. Provisional Application No. 62/705,042, filed Jun. 9, 2020, titled "ENCLOSURE ASSEMBLY FOR ENHANCED COOLING OF DIRECT DRIVE UNIT AND RELATED METHODS," and U.S. Provisional Application No. 62/704,981, filed Jun. 5, 2020, titled "ENCLOSURE ASSEMBLY FOR ENHANCED COOLING OF DIRECT DRIVE UNIT (DDU) AND RELATED METHODS," the disclosures of which are incorporated herein by reference in their entireties.

TECHNICAL FIELD

This disclosure relates to enclosure assemblies and related systems and methods for providing enhanced cooling of a direct drive unit (DDU), such as a direct drive turbine (DDT) connected to a gearbox for driving a driveshaft connected to a pump for use in a hydraulic fracturing systems and methods.

BACKGROUND

During fracturing operations, the equipment onboard fracturing trailers utilizes extensive cooling to facilitate operation throughout the pumping stage. The fracturing pump may have, for example, up to 5% energy loss of energy through heat rejection during operation. Such heat rejection may enter bearings, connecting rods, the casing, clamps and other highly temperature sensitive components in the pumps power end. These components are typically kept lubricated and cooled using lube oil that is pumped continuously through circuits into the pump ensuring that the lube oil is cascaded around the crank case of the fluid pump.

Heat rejection from the pump is still absorbed into the oil, however, and this oil is cooled through a lubrication circuit to ensure that the oil remains at a manageable temperature set out by regulation and/or pump original equipment manufacturers (OEMs). The cooling of oil may be achieved by diverting the oil to a heat exchanger (for example, a fan driven heat exchanger, tube and shell heat exchanger, or other heat exchanger as will be understood by those skilled in the art.) that is sized and configured to be able to

2

remove enough heat from the fluid that will allow the oil to enter the crank case again and absorb more heat rejection.

This cooling cycle may occur constantly onboard fracturing trailers with the operations of the heat exchangers at times being hydraulically or electrically driven. The need for higher power rated fracturing pumps, for example, 5000 HP or 7000 HP rated fracturing pumps, may require larger cooling packages to be able to manage the heat rejection. Accordingly, more heat rejection may directly correlate to the physical footprint of the cooling systems.

SUMMARY OF THE DISCLOSURE

In view of the foregoing, there is an ongoing need for an enclosure assembly and related systems and methods that are more suitable for cooling the DDU of a pumping system, as well as for high-pressure and high-power operations.

Accordingly, it may be seen that a need exists for managing the location of cooling systems to minimize physical footprint, for managing associated power resources efficiently, and for providing effective cooling for fracturing pumps and DDUs. The present disclosure addresses these and other related and unrelated problems in the art.

One exemplary embodiment of the disclosure includes an enclosure assembly to enhance cooling of a hydraulic fracturing direct drive unit (DDU) during operation. An enclosure body may be provided extending at least partially around an enclosure space to house the DDU, which may include a turbine engine that is mechanically connected to a gearbox for driving a driveshaft connected to the gearbox in order to drive a fluid pump. The enclosure assembly may include one or more heat exchanger assemblies connected to the enclosure body for cooling a process fluid associated with one or more of the DDU and the fluid pump, for example, a lubrication or other lubrication medium, and/or a hydraulic/working fluid that is heated during operation. The one or more heat exchanger assemblies may include one or more intake fan assemblies positioned in fluid communication with an external environment surrounding the enclosure body, and one or more intake fan motors may be operatively connected to the one or more intake fan assemblies. Thus, when the one or more intake fan motors is activated, the one or more intake fan assemblies may draw air into the enclosure space from the external environment at the one or more intake fan assemblies and along an airflow path through the enclosure space. One or more radiator assemblies may further be included in the one or more heat exchanger assemblies for receiving the process fluid, and positioned adjacent the one or more intake fan assemblies in the airflow path through the enclosure space to cool the process fluid with air from the external environment as it flows toward the radiator assembly.

In addition, the enclosure assembly may include one or more outlet fan assemblies positioned in fluid communication with the external environment. Accordingly, to maintain a desired temperature of the enclosure space, the one or more outlet fan assemblies may be operatively connected to one or more outlet fan motors to discharge air from the enclosure space to the external environment when the one or more outlet fan motors is activated such that airflow heated by the cooling of the process fluid may be ventilated from the enclosure assembly. The enclosure assembly may also include one or more temperature sensors to detect a temperature of the enclosure space and, further, one or more controllers in electrical communication with the one or more temperature sensors. The one or more controllers may be operatively connected to one or more of the one or more

3

intake fan motors and the one or more outlet fan motors. In this regard, the one or more controllers may activate the respective one or more intake fan motors and the one or more outlet fan motors to rotate the respective one or more intake fan assemblies and the one or more outlet fan assemblies responsive to a predetermined temperature signal from the one or more temperature sensors to discharge heated air from and maintain a desired temperature of the enclosure space.

Another exemplary embodiment of the disclosure includes a fluid pumping system for high-pressure, high-power hydraulic fracturing operations. The system may include a direct drive unit (DDU) having a turbine engine mechanically connected to a gearbox for driving a drive-shaft, and a fluid pump operatively connected to the DDU by the driveshaft for driving the fluid pump. Accordingly, one or more of the DDU and the fluid pump may generate and heat process fluid during operation, which may include lubrication oil or another lubrication medium, and/or a hydraulic or other working fluid. The system may include an enclosure assembly having an enclosure body extending around an enclosure space to house the DDU, and one or more or more heat exchanger assemblies connected to the enclosure body for cooling process fluid associated with one or more of the DDU and the fluid pump. The one or more heat exchanger assemblies of the system may include one or more intake fan assemblies positioned in fluid communication with an external environment surrounding the enclosure body, and one or more intake fan motors may be operatively connected to the one or more intake fan assemblies. When the one or more intake fan motors is activated, the one or more intake fan assemblies may draw air into the enclosure space from the external environment at the one or more intake fan assemblies and along an airflow path through the enclosure space. One or more radiator assemblies may be included in the one or more heat exchanger assemblies for receiving the process fluid, and may be positioned adjacent the one or more intake fan assemblies in the airflow path through the enclosure space to cool the process fluid with the air drawn in from the external environment as it flows through the radiator assembly.

The system's enclosure assembly may also include one or more outlet fan assemblies positioned in fluid communication with the external environment. In order to maintain a desired temperature of the enclosure space, the one or more outlet fan assemblies may be operatively connected to one or more outlet fan motors to discharge air from the enclosure space to the external environment when the one or more outlet fan motors is activated so that airflow in the enclosure space that has been heated from the cooling of the process fluid may be ventilated from the enclosure assembly. The enclosure assembly of the system may also include one or more temperature sensors to detect a temperature of the enclosure space and, further, one or more controllers in electrical communication with the one or more temperature sensors. The one or more controllers may be operatively connected to one or more of the one or more intake fan motors and the one or more outlet fan motors. In this regard, the one or more controllers may activate the respective one or more intake fan motors and the one or more outlet fan motors to rotate the respective one or more intake fan assemblies and the one or more outlet fan assemblies responsive to a predetermined temperature signal from the one or more temperature sensors to discharge heated air from and maintain a desired temperature of the enclosure space

4

Still another exemplary embodiment of the disclosure includes a method of enhancing cooling during operation of a hydraulic fracturing direct drive unit (DDU) having a turbine engine mechanically connected to a gearbox. The method may include operating the DDU to drive a driveshaft operatively connected to a fluid pump such that one or more of the turbine engines and the fluid pump generate and heat process fluid, for example, a lubrication or other lubrication medium, and/or a hydraulic/working fluid. The method may include detecting a temperature in an enclosure space of an enclosure assembly housing the DDU with one or more temperature sensors, and, further, controlling one or more intake fan assemblies of one or more heat exchanger assemblies in the enclosure space to draw air from an external environment into an airflow through the enclosure space based upon a temperature signal detected by the one or more temperature sensors. In this regard, the method may include cooling the process fluid by directing airflow from the one or more intake fan assemblies toward one or more radiator assemblies of the one or more heat exchangers carrying the process fluid. The method may further include controlling one or more outlet fan assemblies to discharge airflow heated by the cooling of the process fluid to the external environment to maintain a desired temperature in the enclosure space.

Those skilled in the art will appreciate the benefits of various additional embodiments reading the following detailed description of the embodiments with reference to the below-listed drawing figures. It is within the scope of the present disclosure that the above-discussed embodiments be provided both individually and in various combinations.

BRIEF DESCRIPTION OF THE DRAWINGS

According to common practice, the various features of the drawings discussed below are not necessarily drawn to scale. Dimensions of various features and elements in the drawings may be expanded or reduced to more clearly illustrate the embodiments of the disclosure.

FIG. 1A is a schematic diagram of a pumping unit according to an embodiment of the disclosure.

FIG. 1B is a schematic diagram of a layout of a fluid pumping system according to an embodiment of the disclosure.

FIG. 2 is a perspective view of an enclosure assembly according to an embodiment of the disclosure.

FIG. 3 is a schematic sectional view of an enclosure body according to an embodiment of the disclosure.

FIG. 4 is a schematic sectional view of an enclosure assembly according to an embodiment of the disclosure.

FIG. 5 is a schematic diagram of a heat exchanger assembly according to an embodiment of the disclosure.

FIG. 6 is a front view of a heat exchanger assembly according to an embodiment of the disclosure.

FIG. 7 is a side view of a heat exchanger assembly according to an embodiment of the disclosure.

FIG. 8 is a plan sectional view of an enclosure assembly according to an embodiment of the disclosure.

FIG. 9 is a side sectional view of an enclosure assembly according to an embodiment of the disclosure.

FIG. 10 is a schematic diagram of a hydraulic circuit according to an embodiment of the disclosure.

FIG. 11 is a schematic diagram of a control circuit according to an embodiment of the disclosure.

Corresponding parts are designated by corresponding reference numbers throughout the drawings.

DETAILED DESCRIPTION

The embodiments of the present disclosure are directed to enclosure assemblies to enhance cooling of a hydraulic fracturing direct drive unit (DDU) during operation. The embodiments of the present disclosure may be directed to such enclosure assemblies for enhanced cooling of DDUs associated with high-pressure, high-power hydraulic fracturing operations.

FIG. 1A illustrates a schematic view of a pumping unit 111 for use in a high-pressure, high power, fluid pumping system 113 (FIG. 1B) for use in hydraulic fracturing operations according to an embodiment of the disclosure. FIG. 1B shows a typical pad layout of the pumping units 111 (indicated as FP1, FP2, FP3, FP4, FP5, FP6, FP7, FP8) with the pumping units all operatively connected to a manifold M that is operatively connected to a wellhead W.

By way of an example, the system 113 is a hydraulic fracturing application that may be sized to achieve a maximum rated horsepower of 24,000 HP for the pumping system 113, including a quantity of eight (8) 3000 horsepower (HP) pumping units 111 that may be used in one embodiment of the disclosure. It will be understood that the fluid pumping system 113 may include associated service equipment such as hoses, connections, and assemblies, among other devices and tools. As shown in FIG. 1A, each of the pumping units 111 are mounted on a trailer 115 for transport and positioning at the jobsite. Each pumping unit 111 includes an enclosure assembly 121 that houses a direct drive unit (DDU) 123 including a gas turbine engine 125 operatively connected to a gearbox 127 or other mechanical transmission.

The pumping unit 111 has a driveshaft 131 operatively connected to the gearbox 127. The pumping unit 111 includes a high-pressure, high-power, reciprocating positive displacement pump 133 that is operatively connected to the DDU 123 via the driveshaft 131. In one embodiment, the pumping unit 111 is mounted on the trailer 115 adjacent the DDU 123.

The trailer 115 includes other associated components such as a turbine exhaust duct 135 operatively connected to the gas turbine engine 125, air intake duct 137 operatively connected to the gas turbine, and other associated equipment hoses, connections, or other components as will be understood by those skilled in the art to facilitate operation of the fluid pumping unit 111.

In the illustrated embodiment, the gas turbine engine 125 may be a Vericor Model TF50F bi-fuel turbine; however, the DDU 123 may include other gas turbines or suitable drive units, systems, and/or mechanisms suitable for use as a hydraulic fracturing pump drive without departing from the disclosure. In one embodiment, the fluid pumping system 113 may include a turbine engine that uses diesel or other fuel as a power source. The gas turbine engine 125 is cantilever mounted to the gearbox 127, with the gearbox 127 supported by the floor of the enclosure assembly 121.

It should also be noted that, while the disclosure primarily describes the systems and mechanisms for use with DDUs 123 to operate fracturing pumping units 111, the disclosed systems and mechanisms may also be directed to other equipment within the well stimulation industry such as, for example, blenders, cementing units, power generators and related equipment, without departing from the scope of the disclosure.

FIGS. 2 and 4 illustrate an enclosure assembly 121 that houses the DDU 123 according to an exemplary embodiment of the disclosure. As shown, the enclosure assembly 121 includes an enclosure body 165 that may extend at least partially around an enclosure space 122 to house one or more portion of the DDU 123 therein. The enclosure space 122 may also be sized and configured to accommodate other DDU/engine equipment, for example, a driveshaft interface, fuel trains, an exhaust system flanged connection, a fire suppression system, bulkheads, exhaust ducting, engine air intake ducting, hydraulic/pneumatic bulkhead hoses, inspection doors/hatches, or other components and equipment as will be understood by those skilled in the art.

In one embodiment, the enclosure body 165 may be a generally box-like or cuboid arrangement of walls, including a first side wall 167, a second side wall 169 opposite the first side wall 167, and an opposing front wall 171 and rear wall 173 each extending from the first side wall 167 to the second side wall 169. The enclosure body 165 may also include a roof/top wall 166 (FIG. 4) and a floor/bottom wall 168. In one embodiment, the floor 168 may be formed of a solid base steel material mounted on a skid structure.

Referring additionally to FIG. 3, one or more of the walls of the enclosure body 165 may be provided with sound-attenuating, e.g., vibration-dampening, properties to minimize the transmission of sound from one or more operations of the DDU 123, e.g., running of the turbine engine 125 and/or the gearbox 127, from the enclosure space 122 to an external environment surrounding the enclosure body 165. In this regard, the walls of the enclosure body 165 may have a configuration in which multiple layers are arranged to provide sound attenuation. Other sound-attenuating features may be incorporated into the construction of the enclosure assembly 121. For example, the gearbox 127 may be provided with shock-absorbing feet or mounts that minimize the transmission of vibrations to the enclosure body 165.

In one embodiment, the walls of the enclosure body 165 may include an outer metallic layer 171, a foam or other polymeric layer 173 and a composite layer 175, and in inner or liner metallic layer 177, with the foam layer 173 and the composite layer 175 positioned between the metallic layers 171, 177.

In one embodiment, the walls 167, 169, 171, 173 of the enclosure body 165 may be formed from approximately 12"x12" panels with an overall thickness of about 4.5" to about 5.25" that may clip, snap, or otherwise connect together in a generally modular arrangement, and the outer metallic layer 171 may be, for example, a 22 ga perforated aluminum sheet, the foam layer 173 may be, for example, a 1" foam layer, the composite layer 175 may be, for example, a 3"-4" layer of mineral wool, and the inner metallic layer 177 may be, for example, perforated 22 ga aluminum. The roof 166 of the enclosure body 165 may have a similar arrangement, with an overall thickness of, for example, about 2" and having the foam layer 173 at a thickness of about, for example, 1.5". The enclosure body 165 may have a different arrangement without departing from the disclosure.

Still referring to FIG. 2, a plurality of doors may be movably connected/attached to the enclosure body 165, e.g., to provide access to the enclosure space 122 for inspections, maintenance, or other operations as will be understood by those skilled in the art. A pair of doors 179 may be hingably connected/attached to the first side wall 167 of the enclosure body 165 to provide access to the enclosure space 122 through openings formed in the first side wall 167 upon movement of the doors 179.

A door **181** may also be movably connected to the second side wall **169** of the enclosure body **165** to provide access to the enclosure space **122** along the second side wall **169**. In one embodiment, the door **181** may be slidably connected/attached to the second side wall **169** on rails, tracks, or other guides as will be understood by those skilled in the art, such that slidable movement of the door **181** exposes an opening in the second side wall **169** through which an operator may access the enclosure space **122**. In one embodiment, the door **181** may have one or more foldable or otherwise reconfigurable portions.

With additional reference to FIG. 4, a generally horizontal partition **183** may extend in general parallel relation with the roof **166** and the floor **168** of the enclosure body **165** so as to provide an upper compartment **185** and a lower compartment **187** of the enclosure space **122**. In one embodiment, the upper compartment **185** may include an air intake assembly that may include an arrangement of ducts, fans, ports, filtration assemblies, blowers, compressors, cooling coils, or other components as will be understood by those skilled in the art, to feed filtered air into the turbine engine **123** positioned in the lower compartment **187**.

In view of the foregoing, the enclosure assembly **121** may be provided with a generally weatherproof or weather-resistant configuration that is sufficiently robust for use in hydraulic fracturing applications, and which additionally provides sound attenuation properties for enclosed and associated equipment. For example, the enclosure assembly **121** may provide sufficient sound attenuation emanating from one or more incorporated heat exchanger assemblies, as described further herein.

During various operations of the pumping unit **133**, e.g., startup and shutdown procedures, idling, maintenance cycles, active driving of the pumping unit **133**, or other operations as will be understood by those skilled in the art, heat may be generated in one or more portions of the pumping unit **133**, for example, via frictional engagement of components of the pumping unit **133** such as pistons, bores, or other components as will be understood by those skilled in the art. In this regard, the pumping unit **133** may employ a fluid heat transfer medium, e.g., a natural or synthetic lubrication oil, to absorb heat from the pumping unit **133** via fluid convection to reduce heat in one or more portions of the DDU **123**.

Similarly, during various operations of the DDU **123**, heat may be generated by one or more portions of the turbine engine **125** and the gearbox **127**. The DDU **123** may thus also employ a fluid heat transfer medium to absorb heat from the DDU **123** via fluid convection to reduce heat in one or more portions of the DDU **123**.

Further, various hydraulic components of the fluid pumping system **113**, e.g., actuators, motors, pumps, blowers, coolers, filters, or other hydraulic components as will be understood by those skilled in the art, that receive pressurized hydraulic fluid or working fluid therethrough may cause such hydraulic fluid/working fluid to increase in temperature during the course of such operation.

The aforementioned fluid heat transfer media, hydraulic fluids/working fluids, and other thermally conductive fluids associated with the fluid pumping system **113** may be collectively referred to as process fluids associated with the respective components of the fluid pumping system **113** herein.

In this regard, the fluid pumping system **113** may include one or more heat exchanger assemblies for cooling/reducing heat in the aforementioned process fluids. Turning to FIG. 5, a heat exchanger assembly **189A** according to an exemplary

embodiment of the disclosure is schematically illustrated. In the illustrated embodiment, the heat exchanger assembly **189A** may be connected to, e.g., attached, mounted, or otherwise supported by, the enclosure body **165**. While the heat exchanger assembly **189A** is illustrated as being positioned in the enclosure space **122**, it will be understood that the heat exchanger assembly **189A** may be connected to the enclosure body **165** and at least partially positioned outside thereof without departing from the disclosure.

Still referring to FIG. 5, the heat exchanger assembly **189A** may include one or more intake fan assemblies **193**, one or more intake fan motors **195** operatively connected to the intake fan assembly **193**, and one or more radiator assemblies **197** positioned adjacent the intake fan assembly **193**. The heat exchanger assembly **189A** may be positioned in alignment with a cutout or opening in the enclosure body **122**, e.g., so that the heat exchanger assembly **189A** may be in at least partial fluid communication with an external environment **E** surrounding the enclosure assembly **121**. In one embodiment, such cutout or opening may be at least partially covered with a mounting plate **194** which may be connected to the heat exchanger assembly **189A**.

A sealing member **198**, for example, a gasket or other polymeric member, may be positioned between the heat exchanger assembly **189A** and the enclosure body **165**, for example, to inhibit the migration or leakage of fluids between the heat exchanger assembly **189A** and the enclosure body **165**.

The one or more intake fan assemblies **193** may include one or more fans **205** (FIG. 6) rotatably connected to the intake fan motor **195** such that, upon receiving a driving signal or other modality of actuation, the intake fan motor **195** rotates the one or more fans **205** to rotate and circulate air through the enclosure space **122**. Such rotatable connection between the intake fan motor **195** and the fan **205** may be a driveshaft, coupling, or other mechanical transmission. The fan **205** may have a plurality of blades/arms for forcing/urging air into an airflow. In this regard, the fan **205** may be provided with blades/arms having a length, pitch, shape, or other features as will be understood by those skilled in the art, configured to influence airflow in a preselected direction.

As shown, the one or more radiator assemblies **197** is positioned adjacent the intake fan assembly **193**. In one embodiment, the radiator assembly **197** may be configured as a tube-and-shell heat exchanger, in which one or more conduits (e.g., tubes, ducts, hoses, fluid lines, or other conduits as will be understood by those skilled in the art) extend along bulkhead fittings on the enclosure body **122** and through an interior of a housing shell **207** to route the process fluid over a sufficient surface area to effect cooling of the process fluid.

The conduits extending through the housing shell **207** may carry process fluid in the form of a fluid heat exchange medium, hydraulic fluid/working fluid, or other fluid. As described further herein, the radiator assembly **197** may be positioned in an airflow path at least partially provided by the intake fan assembly **193** to remove heat from the process fluid running through the conduits. In one embodiment, the radiator assembly **197** may be covered by/positioned adjacent one or more layers of mesh or otherwise porous material.

Referring to FIGS. 6 and 7, the enclosure assembly **121** may include the heat exchanger assembly **189A** (broadly, "low-pressure heat exchanger assembly **189A**") for cooling process fluid received from a low-pressure portion of the fluid pump **133**, and the enclosure assembly **121** may further include a high-pressure heat exchanger assembly **189B** for

cooling process fluid received from a high-pressure portion of the fluid pump 133. The heat exchanger assembly 189B may be similarly configured to the heat exchanger assembly 189A, though the heat exchanger assemblies 189A, 189B may have one or more differences without departing from the disclosure.

As also shown, the heat exchanger assemblies 189A, 189B are supported on a mounting frame 191 with a generally rigid body having outer frame members 199, 200 intersecting at respective joints/plates 201 that may be secured with fasteners such as bolts, screws, rivets, pins, or other fasteners as will be understood by those skilled in the art. As also shown, the mounting frame 191 is provided with one or more flanges or securing tabs 203 extending from one or more of the frame members 199, 200 and that are configured for engagement with the enclosure body 165. In this regard, the securing tabs 203 may have, for example, a generally flat or planar profile and/or may be provided with an opening for receiving a fastener therethrough. In one embodiment, the securing tabs 203 may be integrally formed with one or more of the frame members 199, 200.

The heat exchanger assemblies 189A, 189B may both be connected to the mounting frame 191 in a vertically stacked arrangement, as shown, though each heat exchanger assembly 189A, 189B may be connected to the enclosure body 165 on separate mounting frames without departing from the disclosure. In one embodiment, the mounting frame 191 may be about 0.25" thick, and may be provided with a tolerance of about 0.1" to about 0.2" beyond the boundaries of the heat exchanger assemblies 189A, 189B.

In one embodiment, the mounting frame 191 may be connected to a modular panel of the side wall 167 that is sized and configured to an area larger than that of the heat exchanger assemblies 189A, 189B. In one embodiment, such modular panel may be provided with a tolerance of about 0.35" to about 0.45" beyond the heat exchanger assemblies 189A, 189B.

In one embodiment, and as shown in FIG. 2, the enclosure assembly 121 may include additional or alternative heat exchangers, for example, a heat exchanger 189C for cooling process fluid associated with the turbine engine 125, a heat exchanger 189D for cooling process fluid associated with the gearbox 127, and a heat exchanger 189E for cooling process fluid associated with one or more hydraulic components of the fluid pumping system 113 (e.g., auxiliary/ancillary actuators, pumps, motors, or other hydraulic components as will be understood by those skilled in the art). It will be understood that each of the heat exchanger assemblies 189A, 189B, 189C, 189D, 189E may be sized/scaled/configured according to the process fluids upon which they are operative to cool.

As described herein, the heat exchanger assemblies 189C, 189D, 189E may have a configuration that is substantially similar to that of the heat exchanger assemblies 189A and 189B, though one or more of the heat exchanger assemblies 189A, 189B, 189C, 189D, 189E may have a different configuration without departing from the disclosure. By way of example, two or more of the one or more of the heat exchanger assemblies 189A, 189B, 189C, 189D, 189E may share a common mounting frame, housing shell, intake fan assembly, or other component as will be understood by those skilled in the art.

As shown in FIG. 2, the heat exchanger assemblies 189A, 189B, 189C, 189D, 189E are connected to the enclosure body 165 and positioned in fluid communication with the external environment E such that when the respective intake fan assemblies 193 are driven by the respective intake fan

motors 195, the intake fan assemblies 193 are operative to draw air in from the external environment E toward the respective radiator assemblies 197 to remove heat/cool the process fluids flowing therethrough, and so that they may return to respective portions of the fluid pumping system for continued lubrication/cooling of components of the fluid pumping system 113.

The aforementioned action of the intake fan assemblies 193 causes air from the external environment E to absorb heat from the radiator assemblies 197 as it passes thereby/therethrough and further into the enclosure space 122. In this regard, operation of one or more of the heat exchanger assemblies 189A, 189B, 189C, 189D, 189E may cause an ambient temperature in the enclosure space 122 of the enclosure assembly 121 to increase.

With additional reference to FIGS. 8 and 9, one or more outlet/suction fan assemblies 209 may also be connected to the enclosure body 165. The one or more outlet fan assemblies 209 may have a similar configuration to the aforementioned intake fan assemblies 193, in that they may include one or more outlet fans, e.g., a fan 205, in operative communication with one or more respective motors, e.g., an outlet fan motor 196, such that upon receiving a driving signal or actuation force, the outlet fan motor 196 may drive the fan 205 to rotate. In one embodiment, the outlet fan assembly 209 may include a pair of fans 205 driven by one or more outlet fan motors 196. It will be understood that the one or more inlet fan assemblies 193 and the one or more outlet fan assemblies 209 may be driven by the same motor or combination of motors. Although the one or more outlet fan assemblies 209 has been described herein separately from the heat exchanger assemblies 189A, 189B, 189C, 189D, 189E, it will be understood that one or more of the heat exchanger assemblies 189A, 189B, 189C, 189D, 189E may include the one or more outlet fan assemblies 209 without departing from the disclosure.

In one embodiment, one or more of the heat exchanger assemblies 189A, 189B, 189C, 189D, 189E may be attached to the first side wall 167 of the enclosure body 165, and the outlet fan assembly 209 may be attached to the second side wall 169 of the enclosure body 165. It will be understood that the heat exchanger assemblies 189A, 189B, 189C, 189D, 189E and the outlet fan assembly 209 may be attached to the enclosure body 165 in a different arrangement without departing from the disclosure.

In this regard, upon receipt of an actuation force or driving signal, the one or more outlet fan motors 196 associated with the outlet fan assembly 209 may rotate the fan 205 to discharge air from the enclosure space 122 to the external environment E. Accordingly, the arrangement of the heat exchanger assemblies 189A, 189B, 189C, 189D, 189E and the outlet fan assembly 209 is operative to draw atmospheric/cool air in from the external environment E at the intake fan assembly 193, direct airflow toward the radiator assembly 197 to cool the process fluids flowing there-through, and, further, to ventilate the enclosure assembly 121 by directing an airflow path A from the intake fan assembly 193 to the outlet fan assembly 209 and discharging the air from the enclosure space 122/airflow path A that has been heated from cooling the radiator assembly 197 to the external environment E at the outlet fan assembly 209.

Still referring to FIGS. 8 and 9, in one embodiment, one or more of the heat exchanger assemblies 189A, 189B, 189C, 189D, 189E, in cooperation with the one or more outlet fan assemblies 209, is configured to replace a volume of air in the enclosure space 122 at an interval of about 30 seconds. It will be understood that the heat exchanger

11

assemblies may be configured to replace the same or a different volume of air at a different time interval without departing from the disclosure.

Accordingly, the enclosure assembly **121** may be provided with enhanced cooling capabilities for managing excess heat generated by one or more of the DDU **123**, the fluid pump **113**, and various hydraulic components associated with the fluid pumping system **113**. As described above, one or more of the heat exchanger assemblies **189A**, **189B**, **189C**, **189D**, **189E** is operative to cool process fluid associated with one or more of the DDU **123**, the fluid pump **113**, and various hydraulic components associated with the fluid pumping system **113**. Further, the intake fan assemblies **193** of the heat exchanger assemblies **189A**, **189B**, **189C**, **189D**, **189E** direct the airflow path A through the enclosure space **122** such that, in cooperation with the outlet fan assembly **209**, the air in the enclosure space **122** may be discharged to the external environment E to provide ventilation in the enclosure space **122**. Such ventilation may, for example, maintain a desired temperature of the enclosure space **122**, e.g., to further enhance a temperature differential between the airflow path A and the process fluid in the heat exchanger assemblies **189A**, **189B**, **189C**, **189D**, **189E**.

As described herein, one or more of the motors **195**, **196** may be hydraulic motors, e.g., such that a pressurized working fluid/hydraulic fluid flows therethrough to actuate the motors **195**, **196**.

With additional reference to FIG. **10**, a schematic diagram is provided to show a hydraulic circuit that may be used to drive one or more of the fans **205** of the respective intake fan assemblies **193**. As shown, each intake fan motor **195** includes an inlet port **211** in fluid communication with a hydraulic pump **213** to receive pressurized fluid from the hydraulic pump **213** to actuate the respective intake fan motor **195**. The intake fan motors **195** are also in fluid communication with a return port or outlet port **215** in fluid communication with the hydraulic pump **213** to return hydraulic fluid/working fluid to the respective hydraulic pump **213** after it has passed through/actuated the respective intake fan motor **195**. Each intake fan motor **195** may also include a drain port **217** in fluid communication therewith, for example, to provide drainage of overflow/excess hydraulic fluid/working fluid, to provide a leakage path or pressure release, or other fluid release as will be understood by those skilled in the art. It will be understood that the one or more outlet fan motors **196** may be arranged/controlled in a manner similar to that described above with regard to the inlet fan motors **195**.

It will be understood that the hydraulic pump **213** may be in fluid communication with the respective fluid pump **133**, the turbine engine **125**, the gearbox **127**, and one or more hydraulic components of the fluid pumping system **113** to receive and return process fluid thereto, for example, through an arrangement of fluid lines, manifolds, valves, or other fluid conduit as will be understood by those skilled in the art. In one embodiment, each of the fluid pump **133**, the turbine engine **125**, the gearbox **127**, and one or more hydraulic components of the fluid pumping system **113** may be associated with a separate hydraulic pump **213**, or a combination of hydraulic pumps **213**. In one embodiment, the motors **195** associated with the respective low-pressure portion of the fluid pump **133** and the high-pressure portion of the fluid pump **133** may share one or more common fluid lines.

Each intake fan motor **195** may have an associated solenoid **219** that includes one or more fluid valves to control the flow of hydraulic fluid/working fluid thereto and

12

therefrom. For example, upon receipt of a predetermined electrical signal, each solenoid **219** may actuate, e.g., open or dilate, to permit the flow of hydraulic fluid/working fluid from the hydraulic pump **213** to the respective inlet port **211** and to permit the flow of hydraulic fluid/working fluid from the respective outlet portion **215** to the hydraulic pump **213**. Similarly, the solenoid **219** may close, e.g., restrict or block, the flow of hydraulic fluid/working fluid therethrough upon receipt of a predetermined electrical signal, e.g., a closure signal.

While the intake fan motors **195** described herein have been described as hydraulic motors driven by pressurized hydraulic/working fluid, it will be understood that one or more of the motors **195** (or the motors **196**) may be an electric motor driven by a received electrical actuation/driving signal. In one embodiment, one or more of the motors **195**, **196** may be an electric motor powered from 3-phase electrical power provided by an onboard generator system capable of a voltage output of 480V.

Turning to FIG. **11**, a schematic diagram of a control system that may be used to control the inlet fan motors **195** is illustrated. As shown, each solenoid **219** may be electrically connected to a controller **221**, e.g., a programmable logic controller (PLC), an off-highway multi-controller, a processor-implemented controller, or other control feature as will be understood by those skilled in the art. In this regard, the controller **221** may be operable to actuate the solenoids **219**, e.g., to selectively open and close the valves of the solenoid **219** to permit/restrict the flow of hydraulic fluid/working fluid through the respective inlet fan motors **195**. It will be understood that the one or more outlet fan motors **196** may be controlled in a manner similar to that described above with regard to the inlet fan motors **195**.

In this regard, the controller **221** may be configured to transmit a driving or actuation signal to the respective solenoids **219** upon receipt of a predetermined electrical signal from a thermal/temperature sensor **223** that may be in proximity to the process fluid associated with the respective turbine engine **125**, gearbox **127**, low-pressure portion of the pump **133**, the high-pressure portion of the pump **133**, and one or more hydraulic components of the fluid pumping system **113**. In this regard, one or more temperature sensors **223** may be connected to the enclosure assembly **121** or components thereof. In one embodiment, the sensors **223** may be disposed along a fluid line between the outlet port/return portion **215** of the respective motor **195** and the hydraulic pump **213** and/or a respective reservoir for the process fluid carried therethrough.

In one embodiment, the sensors **223** may be digital thermometers or another electronic sensor that may receive/absorb heat from the associated respective turbine engine **125**, gearbox **127**, low-pressure portion of the pump **133**, the high-pressure portion of the pump **133**, and one or more hydraulic components of the fluid pumping system **113**, and transmit a corresponding electrical signal to the controller **221**. If the respective electrical signal corresponds to a temperature that is at or above a predetermined value or threshold, for example, set by regulation or OEMs, the controller **221** may signal the respective solenoid **219** to open the respective valves.

It will be understood that such actuation of the solenoids **219** may be performed at a constant or predetermined time interval, on-demand, e.g., if and when a predetermined signal is received from the sensors **223**, and/or may be performed proportionally to the temperature of the enclosure space **122**, e.g., so that determining and monitoring greater/lesser temperatures in the enclosure space **122**, the controller

13

221 will proportionally increase/decrease the flow rate of hydraulic/working fluid flowing through the respective intake fan motors 195, and consequently, the speed of the respective associated fans 205.

In one embodiment, one or more of the sensors 223 may include an analog device configured to receive/absorb heat and product a corresponding analog electrical signal without any intermediate processing steps, for example, as in a thermocouple, resistance temperature detector (RTD), or temperature switch. Such analog electrical signal may be a raw value determined by the controller 221 or other processor to correspond to a temperature of the enclosure space 122.

While the hydraulic circuit and control of the respective fans 205 has been described above with regard to the heat exchanger assemblies 189A, 189B, 189C, 189D, 189E, it will be understood that the fans 205 of the outlet fan assembly 209 may be driven and controlled in the same or a similar manner.

Still other embodiments of the disclosure, as shown in FIGS. 1-11, also include methods of enhancing cooling during operation of a hydraulic fracturing direct drive unit (DDU) having a turbine engine mechanically connected to a gearbox. An embodiment of a method may include operating the DDU to drive a driveshaft operatively connected to a fluid pump such that one or more of the turbine engine and the fluid pump generates and heats process fluid, for example, a lubrication or other lubrication medium, and/or a hydraulic/working fluid. The method may include detecting a temperature in an enclosure space of an enclosure assembly housing the DDU with one or more temperature sensors, and, further, controlling one or more intake fan assemblies of one or more heat exchanger assemblies in the enclosure space to draw air from an external environment into an airflow through the enclosure space based upon a temperature signal detected by the one or more temperature sensors. In this regard, the method may include cooling the process fluid by directing airflow from the one or more intake fan assemblies toward one or more radiator assemblies of the one or more heat exchangers carrying the process fluid. The method may further include controlling one or more outlet fan assemblies to discharge airflow heated by the cooling of the process fluid to the external environment to maintain a desired temperature in the enclosure space.

In view of the foregoing, the disclosed embodiments of enclosure assemblies for DDUs may provide for enhanced cooling by the configuration and arrangement of one or more heat exchangers that cool one or more process fluids associated with the DDU and/or an associated fluid pumping system while also providing ventilation and cooling of an enclosure space within the enclosure assembly. In addition to the enhanced cooling of the DDU provided by such an arrangement, the footprint of the enclosure assembly may be minimized and the management of associated power systems may be streamlined.

This is a continuation of U.S. Non-Provisional application Ser. No. 18/096,927, filed Jan. 13, 2023, titled "ENCLOSURE ASSEMBLY FOR ENHANCED COOLING OF DIRECT DRIVE UNIT AND RELATED METHODS," which is a continuation of U.S. Non-Provisional application Ser. No. 17/444,485, filed Aug. 5, 2021, titled "ENCLOSURE ASSEMBLY FOR ENHANCED COOLING OF DIRECT DRIVE UNIT AND RELATED METHODS," now U.S. Pat. No. 11,627,683, issued Apr. 11, 2023, which is a continuation of U.S. Non-Provisional application Ser. No. 17/356,063, filed Jun. 23, 2021, titled "ENCLOSURE ASSEMBLY FOR ENHANCED COOLING OF DIRECT

14

DRIVE UNIT AND RELATED METHODS," now U.S. Pat. No. 11,129,295, issued Sep. 21, 2021, which is a divisional of U.S. Non-Provisional application Ser. No. 17/302,039, filed Apr. 22, 2021, titled "ENCLOSURE ASSEMBLY FOR ENHANCED COOLING OF DIRECT DRIVE UNIT AND RELATED METHODS," now U.S. Pat. No. 11,109,508, issued Aug. 31, 2021, which claims priority to and the benefit of, under 35 U.S.C. § 119(e), U.S. Provisional Application No. 62/705,042, filed Jun. 9, 2020, titled "ENCLOSURE ASSEMBLY FOR ENHANCED COOLING OF DIRECT DRIVE UNIT AND RELATED METHODS," and U.S. Provisional Application No. 62/704,981, filed Jun. 5, 2020, titled "ENCLOSURE ASSEMBLY FOR ENHANCED COOLING OF DIRECT DRIVE UNIT (DDU) AND RELATED METHODS," the disclosures of which are incorporated herein by reference in their entireties.

The foregoing description of the disclosure illustrates and describes various exemplary embodiments. Various additions, modifications, and changes may be made to the exemplary embodiments without departing from the spirit and scope of the disclosure. It is intended that all matter contained in the above description or shown in the accompanying drawings shall be interpreted as illustrative and not in a limiting sense. Additionally, the disclosure shows and describes only selected embodiments of the disclosure, but the disclosure is capable of use in various other combinations, modifications, and environments and is capable of changes or modifications within the scope of the inventive concept as expressed herein, commensurate with the above teachings, and/or within the skill or knowledge of the relevant art. Furthermore, certain features and characteristics of each embodiment may be selectively interchanged and applied to other illustrated and non-illustrated embodiments of the disclosure.

What is claimed is:

1. An enclosure assembly of a hydraulic fracturing power drive unit, the enclosure assembly comprising: an enclosure body at least partially defining an enclosure space to house the hydraulic fracturing power drive unit, the enclosure body comprising: a floor; a plurality of side walls; wherein each of the side walls comprises a plurality of layers, and wherein at least one of the plurality of layers is configured to restrict sound transmission therethrough; wherein the plurality of layers includes: an inner metallic layer; an outer metallic layer; and a foam layer and a wool layer positioned between the inner metallic layer and the outer metallic layer; a roof; and a pair of doors hingably attached to a first side wall of the plurality of side walls; one or more fan assemblies configured to flow air from an external environment surrounding the enclosure body along an airflow path through the enclosure space; a temperature sensor configured to detect a temperature of the enclosure space and output a temperature signal; and a controller in electrical communication with the temperature sensor, the controller configured to activate at least one of the one or more fan assemblies responsive to the temperature signal, thereby to maintain a desired temperature of the enclosure space.

2. The enclosure assembly of claim 1, wherein the one or more fan assemblies includes an intake fan assembly connected to one of the plurality of side walls, the intake fan assembly comprising a motor.

3. The enclosure assembly of claim 1, wherein the one or more fan assemblies includes an outtake fan assembly connected to one of the plurality of side walls, the outtake fan assembly comprising a motor.

15

4. The enclosure assembly of claim 1, wherein the one or more fan assemblies includes an intake fan assembly and an outtake fan assembly, wherein the intake fan assembly is connected to a side wall of the enclosure body and the outtake fan assembly is connected to another, separate side wall of the enclosure body such that the airflow path extends from the intake fan assembly to the outtake fan assembly within the enclosure space.

5. The enclosure assembly of claim 1, wherein the temperature sensor includes: (a) an analog sensor configured to produce an analog electrical signal corresponding to the temperature of the enclosure space, or (b) a digital sensor configured to produce a digital electrical signal corresponding to the temperature of the enclosure space.

6. The enclosure assembly of claim 1, wherein one of the plurality of side walls is configured to receive a driveshaft therethrough.

7. The enclosure assembly of claim 6, wherein the driveshaft is connected to a gearbox disposed within the enclosure body, the gearbox having shock-absorbing mounts configured to reduce transmission of vibrations from the gearbox to the enclosure body.

8. The enclosure assembly of claim 7, further comprising: a gas turbine cantilever mounted to the gearbox and disposed within the enclosure body; and an air intake assembly disposed in the enclosure body in a location between the gas turbine and the roof.

9. A fluid pumping system for hydraulic fracturing operations, the fluid pumping system comprising: a power drive unit, comprising: a gearbox; a gas turbine engine cantilever mounted to the gearbox; and a driveshaft operably connected to the gearbox; a fluid pump operatively connected to the driveshaft; an enclosure assembly including an enclosure body at least partially defining an enclosure space to house at least a portion of the power drive unit, the enclosure body comprising: a floor; a plurality of side walls; wherein each of the side walls comprises a plurality of layers, and wherein at least one of the plurality of layers is configured to restrict sound transmission therethrough; wherein the plurality of layers includes: an inner metallic layer; an outer metallic layer; and a foam layer and a wool layer positioned between the inner metallic layer and the outer metallic layer; a roof; and a pair of doors hingably attached to a first side wall of the plurality of side walls; one or more fan assemblies configured to flow air from an external environment surrounding the enclosure body along an airflow path through the enclosure space; and a temperature sensor configured to detect a temperature of the enclosure space and configured to output a temperature signal.

16

10. The fluid pumping system of claim 9, wherein the one or more fan assemblies includes an intake fan assembly or an outtake fan assembly, wherein the intake fan assembly is connected to a side wall of the enclosure body and the outtake fan assembly is connected to another, separate side wall of the enclosure body such that the airflow path extends from the intake fan assembly to the outtake fan assembly within the enclosure space.

11. The fluid pumping system of claim 9, wherein one of the plurality of side walls is configured to receive the driveshaft therethrough.

12. The fluid pumping system of claim 11, wherein the gearbox is disposed upon shock-absorbing mounts configured to reduce transmission of vibrations from the gearbox to the enclosure body.

13. The fluid pumping system of claim 11, wherein the gearbox is supported by the floor of the enclosure body.

14. A fluid pumping system for hydraulic fracturing operations, the fluid pumping system comprising: a power drive unit, comprising: a gearbox; a gas turbine engine cantilever mounted to the gearbox; and a driveshaft operably connected to the gearbox; a fluid pump operatively connected to the driveshaft; an enclosure assembly including an enclosure body at least partially defining an enclosure space to house at least a portion of the power drive unit, the enclosure body comprising: a floor; a plurality of side walls; wherein each of the side walls comprises a plurality of layers, and wherein at least one of the plurality of layers is configured to restrict sound transmission therethrough; wherein the plurality of layers includes: an inner metallic layer; an outer metallic layer; and a foam layer and a wool layer positioned between the inner metallic layer and the outer metallic layer; a roof; and a pair of doors hingably attached to a first side wall of the plurality of side walls; and one or more fan assemblies configured to flow air from an external environment surrounding the enclosure body along an airflow path through the enclosure space, wherein the power drive unit, the fluid pump, and the enclosure assembly are mounted on a trailer along a longitudinal axis of the driveshaft.

15. The fluid pumping system of claim 14, wherein the gearbox is supported by the floor of the enclosure body.

16. The fluid pumping system of claim 14, wherein the gearbox is disposed upon shock-absorbing mounts configured to reduce transmission of vibrations from the gearbox to the enclosure body.

* * * * *